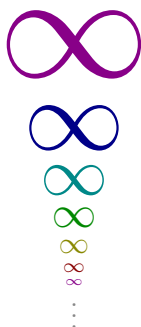


An Infinite Descent into Pure Mathematics

Solution Book



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Chapter 0

Getting started

Exercise 0.1

Think of an example of (a) a true proposition; (b) a false proposition; and (c) a proposition whose truth value you don't know. ◁

❖ Solution

- (a) 2 is an even number.
- (b) 2 is an odd number.
- (c) Every even integer greater than two can be expressed as the sum of two positive prime numbers. [This is *Goldbach's conjecture*, proposed in 1742 by Christian Goldbach. It is a famous example of an extant *open problem* in mathematics. Will you be the one to solve it?] ◁

Exercise 0.2

Suppose that x is a real number that is not an integer. Prove that $\lceil x \rceil = \lfloor x \rfloor + 1$. ◁

❖ Solution

Set $n = \lfloor x \rfloor$. Then $n \leq x < n + 1$ by definition of floor. Since x is not an integer and n is, we have $n < x$; but then $n < x < n + 1$, and so $(n + 1) - 1 < x \leq n + 1$. This inequality demonstrates that $n + 1$ is the ceiling of x , and so $\lceil x \rceil = n + 1 = \lfloor x \rfloor + 1$, as required.

Exercise 0.3

Suppose that x is a real number. Prove that $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$. ◁

❖ Solution

Set $n = \lfloor -x \rfloor$. By definition of floor we have $n \leq -x < n + 1$. Multiplying through by -1 gives $-n \geq x > -(n + 1)$, or equivalently $-n - 1 < x \leq -n$. Thus $-n$ satisfies the definition of being the ceiling of x , so that $\lceil x \rceil = -n = -\lfloor -x \rfloor$. Hence $\lfloor -x \rfloor = -\lceil x \rceil$, as required.

Now set $m = \lceil -x \rceil$. By definition of ceiling we have $m - 1 < -x \leq m$. Multiplying through by -1 and rearranging gives $-m \leq x < -m + 1$. Thus $\lfloor x \rfloor = -m = -\lceil -x \rceil$, and so $\lceil -x \rceil = -\lfloor x \rfloor$, as required.

 Exercise 0.4

Complete the proof of ??.



❖ **Solution**

Let a, b, c, d be integers, with $b \neq 0$ and $d \neq 0$, such that $x = \frac{a}{b}$ and $y = \frac{c}{d}$; we know a, b, c, d exist by the definition of what it means for x and y to be rational. Then:

$$\bullet \quad x - y = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd};$$

$$\bullet \quad xy = \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}; \text{ and}$$

$$\bullet \quad \text{If } y \neq 0, \text{ then } c \neq 0 \text{ (since if } c = 0 \text{ then } y = \frac{c}{d} = \frac{0}{d} = 0), \text{ and so } \frac{x}{y} = \frac{a/b}{c/d} = \frac{ad}{bc}.$$

The expressions on the right-hand side of each of the above chains of equations is a fraction of two integers, with a nonzero integer in the denominator, and so the numbers $x + y$, $x - y$, xy and $\frac{x}{y}$ are all rational.

 Exercise 0.5

Determine the truth values of the following propositions.

(a) $2 \in \mathbb{N}$

(d) $\frac{2}{5} \in \mathbb{Z}$

(f) $\pi^\pi \in \mathbb{Q}$

(b) $-2 \in \mathbb{N}$

(g) $\sqrt{-7} \in \mathbb{R}$

(c) $-2 \in \mathbb{Q}$

(e) $\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} \in \mathbb{Q}$

(h) $\mathbb{N} \in \mathbb{Z}$



❖ **Solution**

- (a) $2 \in \mathbb{N}$ is true, since it is a nonnegative whole number and can be used for counting (for example, I have 2 eyes).
- (b) $-2 \in \mathbb{N}$ is false, since $-2 < 0$ but all natural numbers are nonnegative.
- (c) $-2 \in \mathbb{Q}$ is true, since $-2 = \frac{-2}{1}$, which is an integer divided by a nonzero integer.
- (d) $\frac{2}{5} \in \mathbb{Z}$ is false, since $0 < \frac{2}{5} < 1$, meaning that $\frac{2}{5}$ is strictly between two consecutive integers.
- (e) $\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} \in \mathbb{Q}$ is true: note that $\sqrt{8} = 2\sqrt{2}$, so cancelling the common factor of $\sqrt{2}$ from the numerator and denominator reveals that this number is equal to $\frac{2+1}{2-1} = \frac{3}{1}$, which is an integer divided by a nonzero integer.
- (f) $\pi^\pi \in \mathbb{Q}$. . . OK, sorry, this is (kind of) a trick question. Whether or not π^π is rational is an open problem.
- (g) $\sqrt{-7} \in \mathbb{R}$ is false: the square of every real number is nonnegative, but whatever $\sqrt{-7}$ actually is,^a it presumably satisfies $\sqrt{-7}^2 = -7$, which is negative.
- (h) $\mathbb{N} \in \mathbb{Z}$ might appear true at first sight, since every natural number is indeed an integer. However, $\mathbb{N}$ is *the set of all natural numbers*, which is a set, not an integer, and so the proposition $\mathbb{N} \in \mathbb{Z}$ is false.

^a $\sqrt{-7}$ is an example of a *complex number*, which is a topic for another time.

📎 **Exercise 0.6**

Let $A = \{x \in [10] \mid x \text{ is a power of } 2\}$, let $B = \{2^x \mid x \in [10]\}$ and let $C = \{-99, -97, \dots, 95, 97, 99\}$.

- (a) Find a list notation expression for A .
- (b) Find a list notation expression for B .
- (c) Find a set-builder notation expression for C that involves a set of the form $[n]$ for some $n \in \mathbb{N}$.

◁

❖ **Solution**

(a) The elements of $[10]$ are the natural numbers that are less than 10, and those that are also powers of 2 are $2^0 = 1$, $2^1 = 2$, $2^2 = 4$ and $2^3 = 8$; therefore $A = \{1, 2, 4, 8\}$.

(b) B is obtained by raising 2 to the powers given by the elements of $[10]$; thus

$$B = \{2^0, 2^1, \dots, 2^9\} = \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}$$

(c) An integer is an element of C precisely if its absolute value is odd and less than 100; thus

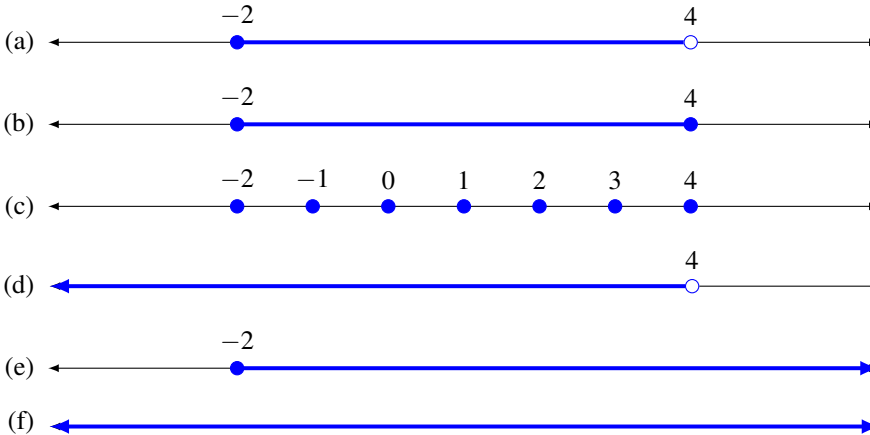
$$C = \{n \in \mathbb{Z} \mid |n| \in [100] \text{ and } n \text{ is odd}\}$$

Another perspective is that by adding 99 to the elements of C and then dividing by 2 we obtain the numbers $0, 1, 2, \dots, 99$, which are the elements of $[100]$; thus we obtain the element of C by reversing these operations, i.e. taking the elements of $[100]$, multiplying them by 2 and then subtracting 99. Thus

$$C = \{2x - 99 \mid x \in [100]\}$$

📎 **Exercise 0.7**

Express each of the sets illustrated below using the most appropriate notation. A filled circle $\bullet$ indicates that a point is included in the set, whereas a hollow circle $\circ$ indicates that an end-point is not included in the interval.



◁

❖ **Solution**

- (a) $[-2, 4)$
- (b) $[-2, 4]$
- (c) $\{-2, -1, 0, 1, 2, 3, 4\}$ — note that interval notation $[-2, 4]$ is *not* appropriate in this case, since the non-integer real numbers between -2 and 5 are not elements of the set.
- (d) $(-\infty, 4)$
- (e) $[-2, \infty)$
- (f) You might be tempted to write $(-\infty, \infty)$ for this one, and that is generally considered acceptable, but it can even more concisely be expressed simply as $\mathbb{R}$.

📎 **Exercise 0.8**

Find the binary, ternary, octal, decimal, hexadecimal and base-36 expansions of the number 21127, using the letters A–F as additional digits for the hexadecimal expansion (representing the numbers 10–15, respectively), and the letters A–Z as additional digits for the base-36 expansion. ◁

📎 **Exercise 0.9**

Prove that 1 divides every integer, and that every integer divides 0. ◁

📎 **Exercise 0.10**

Let $a, b, d \in \mathbb{Z}$. Suppose that d divides a and d divides b . Given integers u and v , prove that d divides $au + bv$. ◁

📎 **Exercise 0.11**

Prove that if an integer a leaves a remainder of r when divided by an integer b , then a leaves a remainder of r when divided by $-b$. ◁

📎 **Exercise 0.12**

Find the base-17 expansion of 408 735 787 and the base-36 expansion of 1 442 151 747. ◁

📎 **Exercise 0.13**

Let r be a rational number and let a be an irrational number. Prove that it is possible that ra be rational, and it is possible that ra be irrational. ◁

📎 **Exercise 0.14**

Interpret the operations of subtraction and division as geometric transformations of the real number line. ◁

Section 0.E

Chapter 0 exercises

0.E.1. The video-sharing website *YouTube* assigns to each video a unique identifier, which is a string of 11 characters from the set

$$\{A, B, \dots, Z, a, b, \dots, z, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, -, _\}$$

This string is actually a natural number expressed in base-64, where the characters in the above set represent the numbers 0 through 63, in the same order—thus C represents 2, c represents 28, 3 represents 55, and _ represents 63. According to this schema, find the natural number whose base-64 expansion is dQw4w9WgXcQ, and find the base-64 expansion of the natural number 7159047702620056984.

0.E.2. Let $a, b, c, d \in \mathbb{Z}$. Under what conditions is $(a + b\sqrt{2})(c + d\sqrt{2})$ an integer?

0.E.3. Suppose an integer m leaves a remainder of i when divided by 3, and an integer n leaves a remainder of j when divided by 3, where $0 \leq i, j < 3$. Prove that $m + n$ and $i + j$ leave the same remainder when divided by 3.

0.E.4. What are the possible remainders of n^2 when divided by 3, where $n \in \mathbb{Z}$?

◆ **Definition 0.E.1**

A set X is **closed** under an operation $\odot$ if, whenever a and b are elements of X , $a \odot b$ is also an element of X .

In **Questions 0.E.5 to 0.E.10**, determine which of the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are closed under the operation $\odot$ defined in the question.

0.E.5. $a \odot b = a + b$

0.E.8. $a \odot b = (a - 1)(b - 1) + 2(a + b)$

0.E.6. $a \odot b = a - b$

0.E.9. $a \odot b = \frac{a}{b^2 + 1}$

0.E.7. $a \odot b = (a - b)(a + b)$

0.E.10. $a \odot b = \frac{a}{\sqrt{b^2 + 1}}$

True–False questions

In **Questions 0.E.11 to 0.E.17**, determine (with proof) whether the statement is true or false.

0.E.11. Every integer is a natural number.

0.E.12. Every integer is a rational number.

0.E.13. Every integer divides zero.

0.E.14. Every integer divides its square.

0.E.15. The square of every rational number is a rational number.

0.E.16. The square root of every positive rational number is a rational number.

0.E.17. When an integer a is divided by a positive integer b , the remainder is always less than a .

Always–Sometimes–Never questions

In Questions 0.E.18 to 0.E.25, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

0.E.18. Let $n, b_1, b_2 \in \mathbb{N}$ with $1 < n < b_1 < b_2$. Then the base- b_1 expansion of n is equal to the base- b_2 expansion of n .

0.E.19. Let $n, b_1, b_2 \in \mathbb{N}$ with $1 < b_1 < b_2 < n$. Then the base- b_1 expansion of n is equal to the base- b_2 expansion of n .

0.E.20. Let $a, b, c \in \mathbb{Z}$ and suppose that a divides c and b divides c . Then ab divides c .

0.E.21. Let $a, b, c \in \mathbb{Z}$ and suppose that a divides c and b divides c . Then ab divides c^2 .

0.E.22. Let $x, y \in \mathbb{Q}$ and let $a, b, c, d \in \mathbb{Z}$ with $cy + d \neq 0$. Then $\frac{ax+b}{cy+d} \in \mathbb{Q}$.

0.E.23. Let $\frac{a}{b}$ be a rational number. Then $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$.

0.E.24. Let $x \in \mathbb{R}$ and assume that $x^2 \in \mathbb{Q}$. Then $x \in \mathbb{Q}$.

0.E.25. Let $x \in \mathbb{R}$ and assume that $x^2 + 1 \in \mathbb{Q}$ and $x^5 + 1 \in \mathbb{Q}$. Then $x \in \mathbb{Q}$.

Part I

Core concepts

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Chapter 1

Logical structure

Section 1.1

Propositional logic

Exercise 1.1.1

Express the proposition ‘John is a mathematician who lives in Pittsburgh’ in the form $p \wedge q$, for propositions p and q . ◁

Exercise 1.1.2

Write a proof tree whose conclusion is the propositional formula $(p \wedge q) \wedge (r \wedge s)$, where p, q, r, s are propositional variables. Express ‘2 is an even prime number and 3 is an odd prime number’ in the form $(p \wedge q) \wedge (r \wedge s)$, for appropriate propositions p, q, r and s , and describe how your proof tree suggests what a proof might look like. ◁

Exercise 1.1.3

Let n be an integer. Prove that n^2 leaves a remainder of 0, 1 or 4 when divided by 5. ◁

Exercise 1.1.4

Let $a, b \in \mathbb{R}$ and suppose $a^2 - 4b \neq 0$. Let α and β be the (distinct) roots of the polynomial $x^2 + ax + b$. Prove that there is a real number c such that either $\alpha - \beta = c$ or $\alpha - \beta = ci$. ◁

Exercise 1.1.5

Let $p(x)$ be a polynomial over $\mathbb{C}$. Prove that if α is a root of $p(x)$, and $a \in \mathbb{C}$, then α is a root of $(x - a)p(x)$. ◁

Exercise 1.1.6

Complete the proof of the quadratic formula. That is, for fixed $a, b \in \mathbb{C}$, prove that if

$$\alpha = \frac{-a + \sqrt{a^2 - 4b}}{2} \quad \text{or} \quad \alpha = \frac{-a - \sqrt{a^2 - 4b}}{2}$$

then α is a root of the polynomial $x^2 + ax + b$. ◁

Exercise 1.1.7

Prove that a natural number n is divisible by 3 if and only if the sum of its base-10 digits is divisible by 3. ◁

 Exercise 1.1.8

Let $x \in \mathbb{R}$. Prove by contradiction that if x is irrational then $-x$ and $\frac{1}{x}$ are irrational. $\triangleleft$

 Exercise 1.1.9

Prove by contradiction that there is no least positive real number. That is, prove that there is not a positive real number a such that $a \leq b$ for all positive real numbers b . $\triangleleft$

 Exercise 1.1.10

Reflect on the proof of ???. Where in the proof did we use the law of excluded middle? Where in the proof did we use proof by contradiction? What was the contradiction in this case? Prove ??? twice more, once using contradiction and not using the law of excluded middle, and once using the law of excluded middle and not using contradiction. $\triangleleft$

 Exercise 1.1.11

Let a and b be irrational numbers. By considering the number $\sqrt{2}^{\sqrt{2}}$, prove that it is possible that a^b be rational. $\triangleleft$

Section 1.2

Variables and quantifiers

 Exercise 1.2.1

Find logical formulae that represent each of the following English statements.

- (a) There is an integer that is divisible by every integer.
- (b) There is no greatest odd integer.
- (c) Between any two distinct rational numbers is a third distinct rational number.
- (d) If an integer has a rational square root, then that root is an integer.

◁

 Exercise 1.2.2

Find statements in plain English, involving as few variables as possible, that are represented by each of the following logical formulae. (The domains of discourse of the free variables are indicated in each case.)

- (a) $\exists q \in \mathbb{Z}, a = qb$ — free variables $a, b \in \mathbb{Z}$
- (b) $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, (b \neq 0 \wedge bx = a)$ — free variable $x \in \mathbb{R}$
- (c) $\forall d \in \mathbb{N}, [(\exists q \in \mathbb{Z}, n = qd) \Rightarrow (d = 1 \vee d = n)]$ — free variable $n \in \mathbb{N}$
- (d) $\forall a \in \mathbb{R}, [a > 0 \Rightarrow \exists b \in \mathbb{R}, (b > 0 \wedge a < b)]$ — no free variables

◁

 Exercise 1.2.3

Prove that every integer is rational.

◁

 Exercise 1.2.4

Prove that every linear polynomial over $\mathbb{Q}$ has a rational root.

◁

 Exercise 1.2.5

Prove that, for all real numbers x and y , if x is irrational, then $x + y$ and $x - y$ are not both rational.

◁

 Exercise 1.2.6

Prove that there is a real number which is irrational but whose square is rational.

◁

 Exercise 1.2.7

Prove that there is an integer which is divisible by zero.

◁

 Exercise 1.2.8

For each of the propositions, write it out as a logical formula involving the $\exists!$ quantifier and then prove it, using the structure of the logical formula as a guide.

- (a) For each real number a , the equation $x^2 + 2ax + a^2 = 0$ has exactly one real solution x .
- (b) There is a unique real number a for which the equation $x^2 + a^2 = 0$ has a real solution x .

(c) There is a unique natural number with exactly one positive divisor.



 Exercise 1.2.9

Let $p(x, y)$ be the statement ' $x + y$ is even'.

- Prove that $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, p(x, y)$ is true.
- Prove that $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, p(x, y)$ is false.



Section 1.3

Logical equivalence

 Exercise 1.3.1

Let p, q and r be propositional variables. Prove that the propositional formula $(p \vee q) \Rightarrow r$ is logically equivalent to $(p \Rightarrow r) \wedge (q \Rightarrow r)$. ◁

 Exercise 1.3.2

Use the definitions of $\wedge$, $\vee$ and $\Rightarrow$ to justify the truth tables in ???. ◁

 Exercise 1.3.3

Use a truth table to prove that $p \Rightarrow q \equiv (\neg p) \vee q$. ◁

 Exercise 1.3.4

Let p, q and r be propositional variables. Use a truth table to prove that the propositional formula $(p \vee q) \Rightarrow r$ is logically equivalent to $(p \Rightarrow r) \wedge (q \Rightarrow r)$. ◁

 Exercise 1.3.5

Use the law of contraposition to prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$, and use the proof technique that this equivalence suggests to prove that an integer is even if and only if its square is even. ◁

 Exercise 1.3.6

Prove that $p \vee q \equiv (\neg p) \Rightarrow q$. Use this logical equivalence to suggest a new strategy for proving propositions of the form $p \vee q$, and use this strategy to prove that if two integers sum to an even number, then either both integers are even or both are odd. ◁

 Exercise 1.3.7

Prove ??(b) thrice: once using the definition of logical equivalence directly (like we did in ???? and [Exercise 1.3.1](#)), once using a truth table, and once using part (a) together with the law of double negation. ◁

 Exercise 1.3.8

Prove that $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ twice, once using a truth table, and once using [Exercise 1.3.3](#) together with de Morgan's laws and the law of double negation. ◁

 Exercise 1.3.9

Prove that $\neg(p \wedge q) \equiv p \Rightarrow (\neg q)$. What strategy does this suggest for proving that a conjunction is false? How does this compare with the strategy suggested by de Morgan's laws for logical operators? ◁

 Exercise 1.3.10

Prove by counterexample that not every rational number can be expressed as $\frac{a}{b}$ where $a \in \mathbb{Z}$ is even and $b \in \mathbb{Z}$ is odd. ◁

 Exercise 1.3.11

Determine which of the following logical formulae are maximally negated.

(a) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, \neg(r(x, y) \wedge s(x, y));$

(b) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, (\neg r(x, y)) \vee (\neg s(x, y));$

(c) $\forall x \in \mathbb{R}, [x > 1 \Rightarrow (\exists y \in \mathbb{R}, [x < y \wedge \neg(x^2 \leq y)]]];$

$$(d) \neg \exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])].$$

◁

 Exercise 1.3.12

Find a maximally negated propositional formula that is logically equivalent to $\neg(p \Leftrightarrow q)$.

◁

 Exercise 1.3.13

Maximally negate the following logical formula, then prove that it is true or prove that it is false.

$$\exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$$

◁

 Exercise 1.3.14

Prove that each of the following is a tautology:

- (a) $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$;
- (b) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$;
- (c) $\exists y \in Y, \forall x \in X, p(x, y) \Rightarrow \forall x \in X, \exists y \in Y, p(x, y)$;
- (d) $[\neg(p \wedge q)] \Leftrightarrow [(\neg p) \vee (\neg q)]$;
- (e) $(\neg \forall x \in X, p(x)) \Leftrightarrow (\exists x \in X, \neg p(x))$.

For each, try to interpret what it means, and how it might be useful in a proof.

◁

Section 1.E

Chapter 1 exercises

1.E.1. For fixed $n \in \mathbb{N}$, let p represent the proposition ‘ n is even’, let q represent the proposition ‘ n is prime’ and let r represent the proposition ‘ $n = 2$ ’. For each of the following propositional formulae, translate it into plain English and determine whether it is true for all $n \in \mathbb{N}$, true for some values of n and false for some values of n , or false for all $n \in \mathbb{N}$.

- (a) $(p \wedge q) \Rightarrow r$
- (b) $q \wedge (\neg r) \Rightarrow (\neg p)$
- (c) $((\neg p) \vee (\neg q)) \vee (\neg r)$
- (d) $(p \wedge q) \wedge (\neg r)$

1.E.2. For each of the following plain English statements, translate it into a symbolic propositional formula. The propositional variables in your formulae should represent the simplest propositions that they can.

- (a) Guinea pigs are quiet, but they’re loud when they’re hungry.
- (b) It doesn’t matter that 2 is even, it’s still a prime number.
- (c) $\sqrt{2}$ can’t be an integer because it is an irrational number.

1.E.3. Let p and q be propositions, and assume that $p \Rightarrow (\neg q)$ is true and that $(\neg q) \Rightarrow p$ is false. Which of the following are true, and which are false?

- (a) q being false is necessary for p to be true.
- (b) q being false is sufficient for p to be true.
- (c) p being true is necessary for q to be false.
- (d) p being true is sufficient for p to be false.

In [Questions 1.E.4 to 1.E.7](#), use the definitions of the logical operators in [Section 1.1](#) to describe what steps should be followed in order to prove the propositional formula in the question; the letters p , q , r and s are propositional variables.

- 1.E.4.** $(p \wedge q) \Rightarrow (\neg r)$
- 1.E.5.** $(p \vee q) \Rightarrow (r \Rightarrow s)$
- 1.E.6.** $(p \Rightarrow q) \Leftrightarrow (\neg p \Rightarrow \neg q)$
- 1.E.7.** $(p \wedge (\neg q)) \vee (q \wedge (\neg p))$

1.E.8. Find a statement in plain English, involving no variables at all, that is equivalent to the logical formula $\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}, (a < b \Rightarrow \exists c \in \mathbb{R}, [a < c < b \wedge \neg(c \in \mathbb{Q})])$. Then prove this statement, using the structure of the logical formula as a guide.

1.E.9. Find a purely symbolic logical formula that is equivalent to the following statement, and then prove it: “No matter which integer you may choose, there will be an integer greater than it.”

1.E.10. Let X be a set and let $p(x)$ be a predicate. Find a logical formula representing the statement ‘there are exactly two elements $x \in X$ such that $p(x)$ is true’. Use the structure of this logical formula to describe how a proof should be structured, and use this structure to prove that there are exactly two real numbers x such that $x^2 = 1$.

1.E.11. Prove that

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$$

How might this logical equivalence help you to prove statements of the form ‘ p if and only if q ’?

1.E.12. Prove using truth tables that $p \Rightarrow q \not\equiv q \Rightarrow p$. Give an example of propositions p and q such that $p \Rightarrow q$ is true but $q \Rightarrow p$ is false.

In **Questions 1.E.13 to 1.E.16**, find a logical formula whose column in a truth table is as shown.

1.E.13.

p	q	
✓	✓	✗
✓	✗	✓
✗	✓	✓
✗	✗	✗

1.E.14.

p	q	
✓	✓	✓
✓	✗	✗
✗	✓	✓
✗	✗	✗

1.E.15.

p	q	r	
✓	✓	✓	✓
✓	✓	✗	✓
✓	✗	✓	✗
✓	✗	✗	✗
✗	✓	✓	✗
✗	✓	✗	✗
✗	✗	✓	✓
✗	✗	✗	✓

1.E.16.

p	q	r	
✓	✓	✓	✓
✓	✓	✗	✗
✓	✗	✓	✗
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	✓	✓
✗	✗	✗	✗

1.E.17. A new logical operator $\uparrow$ is defined by the following rules:

- (i) If a contradiction can be derived from the assumption that p is true, then $p \uparrow q$ is true;
- (ii) If a contradiction can be derived from the assumption that q is true, then $p \uparrow q$ is true;
- (iii) If r is any proposition, and if $p \uparrow q$, p and q are all true, then r is true.

This question explores this curious new logical operator.

- (a) Prove that $p \uparrow p \equiv \neg p$, and deduce that $((p \uparrow p) \uparrow (p \uparrow p)) \equiv p$.
- (b) Prove that $p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q)$ and $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$.
- (c) Find a propositional formula using only the logical operator $\uparrow$ that is equivalent to $p \Rightarrow q$.

1.E.18. Let X be $\mathbb{Z}$ or $\mathbb{Q}$, and define a logical formula p by:

$$\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])$$

Write out $\neg p$ as a maximally negated logical formula. Prove that p is true when $X = \mathbb{Z}$, and p is false when $X = \mathbb{Q}$.

1.E.19. Use ?? to write out a maximally negated logical formula that is equivalent to $\neg \exists! x \in X, p(x)$. Describe the strategy that this equivalence suggests for proving that there is not a unique $x \in X$ such that $p(x)$ is true, and use this strategy to prove that, for all $a \in \mathbb{R}$, if $a \neq -1$ then there is not a unique $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$.

1.E.20. Define a new quantifier $\forall!$ such that de Morgan's laws for quantifiers (??) hold with $\forall$ and $\exists$ replaced by $\forall!$ and $\exists!$, respectively.

True–False questions

In Questions 1.E.21 to 1.E.30, determine (with proof) whether the statement is true or false.

- 1.E.21.** Every implication is logically equivalent to its contrapositive.
- 1.E.22.** Every implication is logically equivalent to its converse.
- 1.E.23.** Every propositional formula whose only logical operators are conjunctions and negations is logically equivalent to a propositional formula whose only logical operators are disjunctions and negations.
- 1.E.24.** Every propositional formula whose only logical operators are conjunctions is logically equivalent to a propositional formula whose only logical operators are disjunctions.
- 1.E.25.** The formulae $p \wedge (q \vee r)$ and $(p \wedge q) \vee r$ are logically equivalent.
- 1.E.26.** The formulae $p \vee (q \vee r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.
- 1.E.27.** The logical formulae $\neg \forall x \in X, \forall y \in Y, p(x, y)$ and $\exists x \in X, \forall y \in Y, p(x, y)$ are logically equivalent.
- 1.E.28.** $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\forall x < 0, \exists y \notin \mathbb{R}, y^2 \neq x$.
- 1.E.29.** $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$.
- 1.E.30.** $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\exists x < 0, \forall y \in \mathbb{R}, y^2 = x$.

Always–Sometimes–Never questions

In Questions 1.E.31 to 1.E.35, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

1.E.31. Let p and q be propositions and assume that p is true. Then $p \Rightarrow q$ is true.

1.E.32. Let p and q be propositions and assume that p is false. Then $p \Rightarrow q$ is true.

1.E.33. Let X and Y be sets and let $p(x, y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \forall y \in Y, p(x, y)$ and $\forall y \in Y, \forall x \in X, p(x, y)$ are logically equivalent.

1.E.34. Let X and Y be sets and let $p(x, y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \exists y \in Y, p(x, y)$ and $\exists y \in Y, \forall x \in X, p(x, y)$ are logically equivalent.

1.E.35. Let X and Y be sets and let $p(x, y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \forall y \in Y, p(x, y)$ and $\exists x \in X, \exists y \in Y, \neg p(x, y)$ are logically equivalent.

Chapter 2

Sets

Section 2.1

Sets

Exercise 2.1.1

Think about why the characterisations of ' $\forall x \in X$ ' in terms of ' $\forall x$ ' and of ' $\exists x \in X$ ' in terms of ' $\exists x$ ' involve different logical operators ($\Rightarrow$ for $\forall$, and $\wedge$ for $\exists$), and verify de Morgan's laws for quantifiers (??) for the expanded forms. $\triangleleft$

❖ Solution

When proving a proposition of the form $\forall x \in X, p(x)$, we let $x \in X$ and then prove $p(x)$ —thus, we introduce a variable x , we assume $x \in X$, and we prove $p(x)$. This is equivalent to introducing an element $x \in \mathcal{U}$ and proving $x \in X \Rightarrow p(x)$, which explains the implication operator in $\forall x, x \in X \Rightarrow p(x)$.

When proving a proposition of the form $\exists x \in X, p(x)$, we define a specific object a , we prove $a \in X$, and we prove $p(a)$. This is equivalent to defining $a \in \mathcal{U}$ and then proving $a \in X \wedge p(a)$, which explains the conjunction operator in $\exists x, x \in X \wedge p(x)$.

We can use maximal negation (as in [Section 1.3](#)) to verify de Morgan's laws:

$$\begin{aligned}
 \neg \forall x \in X, p(x) &\equiv \neg \forall x, (x \in X \Rightarrow p(x)) \\
 &\equiv \exists x, \neg(x \in X \Rightarrow p(x)) && \text{by ??} \\
 &\equiv \exists x, (x \in X \wedge (\neg p(x))) && \text{by Exercise 1.3.8} \\
 &\equiv \exists x \in X, \neg p(x)
 \end{aligned}$$

and

$$\begin{aligned}
 \neg \exists x \in X, p(x) &\equiv \neg \exists x, (x \in X \wedge p(x)) \\
 &\equiv \forall x, \neg(x \in X \wedge p(x)) && \text{by ??} \\
 &\equiv \forall x, (x \in X \Rightarrow (\neg p(x))) && \text{by Exercise 1.3.9} \\
 &\equiv \forall x \in X, \neg p(x)
 \end{aligned}$$

📎 Exercise 2.1.2 (Russell's paradox has been resolved!)

Repeat ?? with the updated definitions of sets and set-builder notation. ◀

❖ Solution

Note that by the new characterisation of set-builder notation, we have $R \in R$ if and only if $R \in \mathcal{U} \wedge R \notin R$.

If $R \in \mathcal{U}$, this means that $R \in R$ if and only if $R \notin R$, which is a contradiction for the same reason as in ??.

It follows that $R \notin \mathcal{U}$; but then all this means is that $R \in R$ is false.

📎 Exercise 2.1.3

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that $[a, b] \subseteq (c, d]$ if and only if $a > c$ and $b \leq d$. ◀

❖ **Solution**

($\Rightarrow$) Assume $[a, b] \subseteq (c, d]$.

- Since $a \in [a, b]$, we have $a \in (c, d]$ and so $a > c$ as required.
- Towards a contradiction, assume $b > d$. Then $\frac{b+d}{2} < \frac{b+b}{2} = b$, so $\frac{b+d}{2} \in [a, b]$, and so $\frac{b+d}{2} \in (c, d]$. This implies that $\frac{b+d}{2} \leq d$; however since $b > d$ we have $\frac{b+d}{2} > \frac{d+d}{2} = d$, contradicting $\frac{b+d}{2} \leq d$. So indeed $b \leq d$, as required.

($\Leftarrow$) Assume $a > c$ and $b \leq d$, and let $x \in [a, b]$, so that $x \in \mathbb{R}$ and $a \leq x < b$. We need to prove that $x \in (c, d]$. Well, combining these inequalities, we have

$$c < a \leq x < b \leq d$$

and so $c < x \leq d$, so that $x \in (c, d]$, as required. □

📎 **Exercise 2.1.4**

Prove that $\{x \in \mathbb{R} \mid x^2 < x\} = (0, 1)$. ◀

❖ **Solution**

Write $A = \{x \in \mathbb{R} \mid x^2 < x\}$; we prove that $A = (0, 1)$ by double containment.

- ($\subseteq$) Let $x \in A$. Then $x \in \mathbb{R}$ and $x^2 < x$, so that $0 < x - x^2 = x(1 - x)$. Now, if $x \leq 0$ then $1 - x > 0$, and so $x(1 - x) \leq 0$; and if $x \geq 1$ then $x > 0$ and $1 - x \leq 0$, and so $x(1 - x) \leq 0$. Thus we must have $0 < x < 1$, and so $x \in (0, 1)$, as required.
- ($\supseteq$) Let $x \in (0, 1)$. Then $x \in \mathbb{R}$ and $0 < x < 1$. Since $x > 0$, the inequality $x < 1$ is preserved when we multiply through by x , and so $x^2 \leq x$. Hence $x \in A$, as required.

By double containment, we have $A = (0, 1)$. □

📎 **Exercise 2.1.5**

Prove by double containment that $\{0, 1\} = \{1, 0\}$ and $\{0, 0\} = \{0\}$. ◀

❖ Solution

Proving $\{0, 1\} = \{1, 0\}$:

- ($\subseteq$) Let $x \in \{0, 1\}$. Then either $x = 0 \in \{1, 0\}$ or $x = 1 \in \{1, 0\}$; in both cases we see that $x \in \{1, 0\}$.
- ($\supseteq$) Identical to ($\subseteq$), but swap the roles of 0 and 1.

Hence $\{0, 1\} = \{1, 0\}$ by double containment. □

Proving $\{0, 0\} = \{0\}$:

- ($\subseteq$) Let $x \in \{0, 0\}$. Then $x = 0$ or $x = 0 \dots$ so $x = 0 \in \{0\}$.
- ($\supseteq$) Let $x \in \{0\}$. Then $x = 0 \in \{0, 0\}$.

Hence $\{0, 0\} = \{0\}$ by double containment. □

These proofs are a little bit silly, but they demonstrate that in list notation for sets, we can change the order that the elements are listed and we can repeat elements in the list without changing the set.

📎 Exercise 2.1.6

Let $a, b \in \mathbb{R}$. Prove that $[a, b]$ is empty if and only if $a > b$, and that (a, b) is empty if and only if $a \geq b$. ◁

📎 Exercise 2.1.7

Let E be an empty set and let $p(x)$ be a predicate with one free variable x with domain of discourse E . Show that the proposition $\forall x \in E, p(x)$ is true, and that the proposition $\exists x \in E, p(x)$ is false. What does the proposition $\forall x \in E, x \neq x$ mean in English? Is it true? ◁

📎 Exercise 2.1.8

Is it true or false that $\emptyset \in X$ for all sets X ? Is it true or false that $\emptyset \subseteq X$ for all sets X ? ◁

📎 Exercise 2.1.9

Explain why the sets $\emptyset$, $\{\emptyset\}$ and $\{\{\emptyset\}\}$ are different, and then determine the truth values of each of the following propositions:

- | | | |
|-------------------------------------|---------------------------------------|---|
| (a) $\emptyset \in \emptyset$ | (c) $\emptyset \in \{\emptyset\}$ | (e) $\{\emptyset\} \subseteq \{\emptyset\}$ |
| (b) $\emptyset \subseteq \emptyset$ | (d) $\emptyset \in \{\{\emptyset\}\}$ | (f) $\{\emptyset\} \subseteq \{\{\emptyset\}\}$ |

Find more examples of simple elementhood and subsethood propositions—that is, propositions of the form $A \in B$ or $A \subseteq B$ —involving the three sets $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$; you should find at least one true and one false elementhood proposition, and at least one true and one false subsethood proposition. ◁

📎 Exercise 2.1.10

Write out the elements of $\mathcal{P}(\{1, 2, 3\})$. ◁

📎 Exercise 2.1.11

Let X be a set. Show that $\emptyset \in \mathcal{P}(X)$ and $X \in \mathcal{P}(X)$. ◁

📎 Exercise 2.1.12

Write out the elements of $\mathcal{P}(\emptyset)$, $\mathcal{P}(\mathcal{P}(\emptyset))$ and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$. ◁

 Exercise 2.1.13

Determine, with proof, whether or not each of the following statements is true.

- (a) $\mathcal{P}(\emptyset) \in \{\emptyset\}$;
- (b) $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$;
- (c) $\mathcal{P}(\mathcal{P}(\emptyset)) \in \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$.

Repeat the exercise with all instances of ' $\in$ ' replaced by ' $\subseteq$ '.

◁

Section 2.2

Set operations

 Exercise 2.2.1

Prove that $[0, \infty) \cap \mathbb{Z} = \mathbb{N}$.

 **Exercise 2.2.2**

Write down the elements of the set $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\}$.

 **Exercise 2.2.3**

Express $[-2, 5) \cap [4, 7)$ as a single interval.

 **Exercise 2.2.4**

Let $A \subseteq \mathbb{R}$. Find expressions using the intersection operation for (a) the set of all positive elements of A , and (b) the set of all integer elements of A .

 **Exercise 2.2.5**

Let X be a set. Prove that $X \cap \emptyset = \emptyset$ and that $X \cap X = X$.

 **Exercise 2.2.6**

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that the open intervals (a, b) and (c, d) are disjoint if and only if $b \leq c$ or $d \leq a$.

 **Exercise 2.2.7**

Let $A \subseteq \mathbb{R}$. Find an expression involving the intersection operation for the set of all non-negative rational elements of A .

 **Exercise 2.2.8**

For each $n \in \mathbb{Z}$, let $A_n = \{x \in \mathbb{R} \mid x \neq n\}$. Describe the set $\bigcap_{n \in \mathbb{Z}} A_n$ in plain terms.

 **Exercise 2.2.9**

Write down the elements of the set

$$\{0, 1, 4, 7\} \cup \{1, 2, 3, 4, 5\}$$

 **Exercise 2.2.10**

Express $[-2, 5) \cup [4, 7)$ as a single interval.



The union operation allows us to define the following class of sets that will be particularly useful for us when studying counting principles in [Section 8.1](#).

 Exercise 2.2.11

Let X and Y be sets. Prove that $X \subseteq Y$ if and only if $X \cup Y = Y$.

 **Exercise 2.2.12 (Distributivity of union over intersection)**

Let X, Y, Z be sets. Prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.

 **Exercise 2.2.13**

Define $X = \{x \in \mathbb{R} \mid x^4 - 5x^2 + 4 > 0\}$. Find an expression for X as a union of intervals.

 **Exercise 2.2.14**

State and prove distributivity laws for general finitary unions and intersections of sets, generalising ?? and Exercise 2.2.12. ◁

 Exercise 2.2.15

Express $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}]$ as an interval. ◁

 Exercise 2.2.16

Prove that $\bigcap_{n \in \mathbb{N}} [n] = \emptyset$ and $\bigcup_{n \in \mathbb{N}} [n] = \mathbb{N}$. ◁

 Exercise 2.2.17

Find a family of sets $\{X_n \mid n \in \mathbb{N}\}$ such that all three of the following properties hold: (i) $\bigcup_{n \in \mathbb{N}} X_n = \mathbb{N}$;

(ii) $\bigcap_{n \in \mathbb{N}} X_n = \emptyset$; and (iii) $X_i \cap X_j \neq \emptyset$ for all $i, j \in \mathbb{N}$. ◁

 Exercise 2.2.18

Write down the elements of the set

$$\{0, 1, 4, 7\} \setminus \{1, 2, 3, 4, 5\}$$

 Exercise 2.2.19

Express $[-2, 5) \setminus [4, 7)$ and $[4, 7) \setminus [-2, 5)$ as intervals. ◁

 Exercise 2.2.20

Let A be the set of all real numbers that are not integers.

(a) Find an expression for A as a relative complement of one set in another.

(b) Find an expression for A as a union of intervals. ◁

 Exercise 2.2.21

Let X and A be sets. Prove that $X \setminus (X \setminus A) = X \cap A$, and deduce that $A \subseteq X$ if and only if $X \setminus (X \setminus A) = A$. ◁

 Exercise 2.2.22

Complete the proof of de Morgan's laws for sets. ◁

 Exercise 2.2.23

Write down all ordered lists that have all of the elements of $\{1, 2, 3\}$ as terms and have no repeated terms. ◁

 Exercise 2.2.24

Write down the elements of the set $\{1, 2\} \times \{3, 4, 5\}$. ◁

 Exercise 2.2.25

Let X be a set. Prove that $X \times \emptyset = \emptyset$. ◁

 Exercise 2.2.26

Let X , Y and Z be sets. Under what conditions is it true that $X \times Y = Y \times X$? ◁

 Exercise 2.2.27

Consider the following sets:

$$\mathbb{R}^3, \mathbb{R}^2 \times \mathbb{R}, \mathbb{R} \times \mathbb{R}^2, \mathbb{R} \times \mathbb{R} \times \mathbb{R}, (\mathbb{R} \times \mathbb{R}) \times \mathbb{R}, \mathbb{R} \times (\mathbb{R} \times \mathbb{R})$$

Are all of these sets equal? If not, which pairs are (and which are not) equal to each other, and is there another sense in which they are all equivalent? $\triangleleft$

Section 2.E

Chapter 2 exercises

Sets

2.E.1. Express the following sets in the indicated form of notation.

- (a) $\{n \in \mathbb{Z} \mid n^2 < 20\}$ in list notation;
- (b) $\{4k + 3 \mid k \in \mathbb{N}\}$ in implied list notation;
- (c) The set of all odd multiples of six in set-builder notation;
- (d) The set $\{1, 2, 5, 10, 17, \dots, n^2 + 1, \dots\}$ in set-builder notation.

2.E.2. Find sets X_n for each $n \in \mathbb{N}$ such that $X_{n+1} \subsetneq X_n$ for all $n \in \mathbb{N}$. Can any of the sets X_n be empty?

2.E.3. Express the set $\mathcal{P}(\{\emptyset, \{\emptyset, \{\emptyset\}\})$ in list notation.

2.E.4. Let X be a set and let $U, V \subseteq X$. Prove that U and V are disjoint if and only if $U \subseteq X \setminus V$.

2.E.5. For each of the following statements, determine whether or not it is true for all sets A and X , and prove your claim.

- (a) If $X \setminus A = \emptyset$, then $X = A$.
- (b) If $X \setminus A = X$, then $A = \emptyset$.
- (c) If $X \setminus A = A$, then $A = \emptyset$.
- (d) $X \setminus (X \setminus A) = A$.

2.E.6. For each of the following statements, determine whether it is true for all sets X, Y , false for all sets X, Y , or true for some choices of X and Y and false for others.

- (a) $\mathcal{P}(X \cup Y) = \mathcal{P}(X) \cup \mathcal{P}(Y)$
- (b) $\mathcal{P}(X \cap Y) = \mathcal{P}(X) \cap \mathcal{P}(Y)$
- (c) $\mathcal{P}(X \times Y) = \mathcal{P}(X) \times \mathcal{P}(Y)$
- (d) $\mathcal{P}(X \setminus Y) = \mathcal{P}(X) \setminus \mathcal{P}(Y)$

2.E.7. Let F be a set whose elements are all sets. Prove that if $\forall A \in F, \forall x \in A, x \in F$, then $F \subseteq \mathcal{P}(F)$.

Questions 2.E.8 to 2.E.13 concern the *symmetric difference* of sets, defined below.

◆ **Definition 2.E.1**

The **symmetric difference** of sets X and Y is the set $X \triangle Y$ ([LaTeX code: \triangle](#)) defined by

$$X \triangle Y = \{a \mid a \in X \text{ or } a \in Y \text{ but not both}\}$$

2.E.8. Prove that $X \triangle Y = (X \setminus Y) \cup (Y \setminus X) = (X \cup Y) \setminus (X \cap Y)$ for all sets X and Y .

2.E.9. Let X be a set. Prove that $X \triangle X = \emptyset$ and $X \triangle \emptyset = X$.

2.E.10. Let X and Y be sets. Prove that $X = Y$ if and only if $X \triangle Y = \emptyset$.

2.E.11. Prove that sets X and Y are disjoint if and only if $X \triangle Y = X \cup Y$.

2.E.12. Prove that $X \triangle (Y \triangle Z) = (X \triangle Y) \triangle Z$ for all sets X, Y and Z .

2.E.13. Prove that $X \cap (Y \triangle Z) = (X \cap Y) \triangle (X \cap Z)$ for all sets X, Y and Z .

◆ **Definition 2.E.2**

A subset $U \subseteq \mathbb{R}$ is **open** if, for all $a \in U$, there exists $\delta > 0$ such that $(a - \delta, a + \delta) \subseteq U$.

In Questions [Questions 2.E.14](#) to [2.E.17](#) you will prove some elementary facts about open subsets of $\mathbb{R}$.

2.E.14. For each of the following subsets of $\mathbb{R}$, determine (with proof) whether it is open:

- | | | |
|-------------------|--------------------|---|
| (a) $\emptyset$; | (c) $(0, 1]$; | (e) $\mathbb{R} \setminus \mathbb{Z}$; |
| (b) $(0, 1)$; | (d) $\mathbb{Z}$; | (f) $\mathbb{Q}$. |

2.E.15. Prove that a subset $U \subseteq \mathbb{R}$ is open if and only if, for all $a \in U$, there exist $u, v \in \mathbb{R}$ such that $u < a < v$ and $(u, v) \subseteq U$.

2.E.16. In this question you will prove that the intersection of finitely many open sets is open, but the intersection of infinitely many open sets might not be open.

- (a) Let $n \geq 1$ and suppose $U_1, U_2, \dots, U_n$ are open subsets of $\mathbb{R}$. Prove that the intersection $U_1 \cap U_2 \cap \dots \cap U_n$ is open.
- (b) Prove that $(0, 1 + \frac{1}{n})$ is open for all $n \geq 1$, but that $\bigcap_{n \geq 1} (0, 1 + \frac{1}{n})$ is not open.

2.E.17. Prove that a subset $U \subseteq \mathbb{R}$ is open if and only if it can be expressed as a union of open intervals—more precisely, $U \subseteq \mathbb{R}$ is open if and only if, for some indexing set I , there exist real numbers a_i, b_i for each $i \in I$, such that $U = \bigcup_{i \in I} (a_i, b_i)$.

2.E.18. Let $\{A_i \mid i \in \mathbb{N}\}$ and $\{B_j \mid j \in \mathbb{N}\}$ be families of sets such that:

- (i) For all $i \in \mathbb{N}$, there exists some $j \geq i$ such that $B_j \subseteq A_i$; and
- (ii) For all $j \in \mathbb{N}$, there exists some $i \geq j$ such that $A_i \subseteq B_j$.

Prove that $\bigcap_{i \in \mathbb{N}} A_i = \bigcap_{j \in \mathbb{N}} B_j$.

True–False questions

In [Questions 2.E.19](#) to [2.E.24](#), determine (with proof) whether the statement is true or false.

2.E.19. If E is a set and $\neg \forall x, x \in E$, then $E = \emptyset$.

2.E.20. If E is a set and $\forall x, \neg x \in E$, then $E = \emptyset$.

2.E.21. $\{1, 2, 3\} = \{1, 2, 1, 3, 2, 1\}$

2.E.22. $\emptyset \in \emptyset$

2.E.23. $\emptyset \in \{\emptyset\}$

2.E.24. $\emptyset \in \{\{\emptyset\}\}$.

Always–Sometimes–Never questions

In Questions 2.E.25 to 2.E.40, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

2.E.25. Let a and b be arbitrary objects. Then $\{a, b\} = \{b, a\}$.

2.E.26. Let a, b and c be arbitrary objects. Then $\{a, b\} = \{a, c\}$.

2.E.27. Let X and Y be sets and suppose there is some a such that $a \in X \Leftrightarrow a \in Y$. Then $X = Y$.

2.E.28. Let X and Y be sets and suppose there is some a such that $a \in X$ but $a \notin Y$. Then $X = Y$.

2.E.29. Let X and Y be sets with $X \neq Y$. Then $\forall a, a \in X \Rightarrow a \notin Y$.

2.E.30. Let X and Y be sets with $X \cap Y = \emptyset$. Then $\forall a, a \in X \Rightarrow a \notin Y$.

2.E.31. Let X and Y be sets with $X \neq Y$. Then $\exists a, a \in X \wedge a \notin Y$.

2.E.32. Let X and Y be sets of sets and suppose that $X \in Y$. Then $X \subseteq Y$.

2.E.33. Let X be a set. Then $\emptyset \in \mathcal{P}(X)$.

2.E.34. Let X be a set. Then $\{\emptyset\} \in \mathcal{P}(X)$.

2.E.35. Let X be a set. Then $\{\emptyset, X\} \subseteq \mathcal{P}(X)$.

2.E.36. Let X and Y be sets such that $\mathcal{P}(X) = \mathcal{P}(Y)$. Then $X = Y$.

2.E.37. Let X and Y be sets such that $X \setminus Y = \emptyset$. Then $X = Y$.

2.E.38. Let X, Y and Z be sets such that $X \setminus Y = Z$ and $Y \subseteq Z$. Then $X = Y \cup Z$.

2.E.39. Let X and Y be sets. Then $X \cup Y = X \cap Y$.

2.E.40. Let X be a set and let $A \subseteq X$. Then $A \in X$.

Chapter 3

Functions

Section 3.1

Functions

Exercise 3.1.1

Recall [Exercise 1.2.8](#). Which of the statements (a), (b) or (c) is of the form $\forall x \in X, \exists! y \in Y, p(x, y)$? For each statement of this form, determine the domain and codomain of the corresponding function, and write an expression defining this function. 

Exercise 3.1.2

Find a function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ whose graph is equal to the set

$$\{\dots, (-2, -5), (-1, -2), (0, 1), (1, 4), (2, 7), (3, 10), \dots\}$$



Exercise 3.1.3

For each of the following specifications of sets X, Y, G , determine whether or not G is the graph of a function from X to Y .

- (a) $X = \mathbb{R}, Y = \mathbb{R}, G = \{(a, a^2) \mid a \in \mathbb{R}\}$;
- (b) $X = \mathbb{R}, Y = \mathbb{R}, G = \{(a^2, a) \mid a \in \mathbb{R}\}$;
- (c) $X = [0, \infty), Y = [0, \infty), G = \{(a^2, a) \mid a \in [0, \infty)\}$;
- (d) $X = \mathbb{Q}, Y = \mathbb{Q}, G = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid xy = 1\}$.
- (e) $X = \mathbb{Q}, Y = \mathbb{Q}, G = \{(a, a) \mid a \in \mathbb{Z}\}$;



Exercise 3.1.4

Which of the following specifications of functions are well-defined?

- (a) $g : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by the equation $(x+1)g(x) = 1$ for all $x \in \mathbb{Q}$;

(b) $h : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $(x+1)h(x) = 1$ for all $x \in \mathbb{N}$;

(c) $k : \mathbb{N} \rightarrow \mathbb{N}$ defined by $(x+1)k(x) = 1$ for all $x \in \mathbb{N}$;

(d) $\ell : \mathbb{N} \rightarrow \mathbb{N}$ defined by $\ell(x) = \ell(x)$ for all $x \in \mathbb{N}$.

◁

 Exercise 3.1.5

Find a condition on sets X and Y such that the specification of a function $i : X \cup Y \rightarrow \{0, 1\}$ given by

$$i(z) = \begin{cases} 0 & \text{if } z \in X \\ 1 & \text{if } z \in Y \end{cases}$$

to be well-defined.

◁

 Exercise 3.1.6

Let $f, g, h, k : \mathbb{Q} \rightarrow \mathbb{Q}$ be as in ???. Compute equations defining the following composites:

(a) $f \circ g$;

(b) $g \circ f$;

(c) $((f \circ g) \circ h) \circ k$;

(d) $f \circ (g \circ (h \circ k))$;

(e) $(g \circ g) \circ (g \circ g)$.

◁

 Exercise 3.1.7

Prove that composition of functions is *associative*, that is, if $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow W$ are functions, then

$$h \circ (g \circ f) = (h \circ g) \circ f : X \rightarrow W$$

As a consequence of associativity, when we want to compose more than two functions, it doesn't matter what order we compose the functions in. As such, we can just write $h \circ g \circ f$.

◁

 Exercise 3.1.8

Let $f : X \rightarrow Y$ and $g : Z \rightarrow W$ be functions, and suppose that $Y \subsetneq Z$. Note that there is a function $h : X \rightarrow W$ defined by $h(x) = g(f(x))$ for all $x \in X$. Write h as a composite of functions involving f and g .

◁

 Exercise 3.1.9

Prove parts (b) and (c) of ??.

◁

 Exercise 3.1.10

Use characteristic functions to prove de Morgan's laws for pairwise unions and intersections (??).

◁

 Exercise 3.1.11

For each of the following functions f and subsets U of their domain, describe the image $f[U]$.

(a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 3n$, with $U = \mathbb{N}$;

(b) $f : X \rightarrow X \times X$ (where X is any set) defined by $f(x) = (x, x)$ with $U = X$;

(c) $f : \{a, b, c\} \rightarrow \{1, 2, 3\}$ defined by $f(a) = 1$, $f(b) = 3$ and $f(c) = 1$, with $U = \{a, b, c\}$.

◁

 Exercise 3.1.12

Prove that $f[\emptyset] = \emptyset$ for all functions f .

◁

 Exercise 3.1.13

Let $f : X \rightarrow Y$ be a function and let $U, V \subseteq X$. We saw in ?? that $f[U \cap V] \subseteq f[U] \cap f[V]$. Determine which of the following is true, and for each, provide a proof of its truth or falsity:

(a) $f[U] \cap f[V] \subseteq f[U \cap V]$;

(b) $f[U \cup V] \subseteq f[U] \cup f[V]$;

(c) $f[U] \cup f[V] \subseteq f[U \cup V]$.

◁

 Exercise 3.1.14

Let $f : X \rightarrow Y$ be a function. Prove that $f^{-1}[\emptyset] = \emptyset$ and $f^{-1}[Y] = X$.

◁

 Exercise 3.1.15

Let X be a set. Prove that every function $f : X \rightarrow \{0, 1\}$ is the characteristic function of the subset $f^{-1}[\{1\}] \subseteq X$.

◁

Section 3.2

Injections and surjections

 Exercise 3.2.1

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. Prove that if $g \circ f$ is injective, then f is injective. ◁

 Exercise 3.2.2

Write out what it means to say a function $f : X \rightarrow Y$ is *not* injective, and say how you would prove that a given function is not injective. Give an example of a function which is not injective, and use your proof technique to write a proof that it is not injective. ◁

 Exercise 3.2.3

For each of the following functions, determine whether it is injective or not injective.

- $f : \mathbb{N} \rightarrow \mathbb{Z}$, defined by $f(n) = n^2$ for all $n \in \mathbb{N}$.
 - $g : \mathbb{Z} \rightarrow \mathbb{N}$, defined by $g(n) = n^2$ for all $n \in \mathbb{Z}$.
 - $h : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, defined by $h(x, y, z) = 2^x \cdot 3^y \cdot 5^z$ for all $x, y, z \in \mathbb{N}$.
- ◁

 Exercise 3.2.4

Let $a, b \in \mathbb{R}$ with $b \neq 0$, and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) = a + bt$ for all $t \in \mathbb{R}$. Prove that f is injective. ◁

 Exercise 3.2.5

For each of the following pairs of sets (X, Y) , determine whether the function $f : X \rightarrow Y$ defined by $f(x) = 2x + 1$ is surjective.

- (a) $X = \mathbb{Z}$ and $Y = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$;
 - (b) $X = \mathbb{Z}$ and $Y = \mathbb{Z}$;
 - (c) $X = \mathbb{Q}$ and $Y = \mathbb{Q}$;
 - (d) $X = \mathbb{R}$ and $Y = \mathbb{R}$.
- ◁

 Exercise 3.2.6

Let $f : X \rightarrow Y$ be a function. Find a subset $V \subseteq Y$ and a surjection $g : X \rightarrow V$ agreeing with f (that is, such that $g(x) = f(x)$ for all $x \in X$). ◁

 Exercise 3.2.7

Let $f : X \rightarrow Y$ be a function. Prove that f is surjective if and only if $Y = f[X]$ ◁

 Exercise 3.2.8

Let $f : X \rightarrow Y$ be a function. Prove that there is a set Z and functions

$$p : X \rightarrow Z \quad \text{and} \quad i : Z \rightarrow Y$$

such that p is surjective, i is injective, and $f = i \circ p$. ◁

 Exercise 3.2.9

Let $f : X \rightarrow \mathcal{P}(X)$ be a function. By considering the set $B = \{x \in X \mid x \notin f(x)\}$, prove that f is not surjective. $\triangleleft$

 Exercise 3.2.10

Let X be a set. Prove that the identity function $\text{id}_X : X \rightarrow X$ is a bijection. $\triangleleft$

✦ Solution

- **(Injectivity)** Let $a, b \in X$ and assume that $\text{id}_X(a) = \text{id}_X(b)$. Then $a = b$ by definition of id_X , as required.
- **(Surjectivity)** Let $x \in X$. Then $x = \text{id}_X(x)$ by definition of id_X , as required.

Since id_X is both injective and surjective, it is bijective. $\square$

 Exercise 3.2.11

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f$ is a bijection. $\triangleleft$

 Exercise 3.2.12

Let $f : X \rightarrow Y$ be a function. Prove that f is injective if and only if

$$\forall y \in f[X], \exists! x \in X, y = f(x)$$

 $\triangleleft$ **Exercise 3.2.13**

Define a function $e : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by $e(m, n) = 2^m \cdot 3^n$. Prove that e is injective. We can think of e as encoding *pairs* of natural numbers as single natural numbers—for example, the pair $(4, 1)$ is encoded as $2^4 \cdot 3^1 = 48$. For each of the following natural numbers k , find the pairs of natural numbers encoded by e as k :

$$1 \quad 24 \quad 7776 \quad 59049 \quad 396718580736$$

 $\triangleleft$ **Exercise 3.2.14**

Let $f : X \rightarrow Y$ be a function. Prove that if f has a left inverse, then f is injective. $\triangleleft$

 Exercise 3.2.15

Let $f : X \rightarrow Y$ be a function with left inverse $g : Y \rightarrow X$. Prove that g is a surjection. $\triangleleft$

 Exercise 3.2.16

Let $f : X \rightarrow Y$ be a function. Prove that if f has a right inverse, then f is surjective. $\triangleleft$

 Exercise 3.2.17

The following functions have two-sided inverses. For each, find its inverse and prove that it is indeed an inverse.

(a) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x+1}{3}$ for all $x \in \mathbb{R}$;

(b) $g : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N})$ defined by $g(X) = \mathbb{N} \setminus X$ for all $X \in \mathcal{P}(\mathbb{N})$;

(c) $h : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined by $h(m, n) = 2^m(2n+1) - 1$ for all $m, n \in \mathbb{N}$.

<

 Exercise 3.2.18

Let $f : X \rightarrow Y$ be a function. Then f is bijective if and only if f has an inverse.

<

 Exercise 3.2.19

Let $f : X \rightarrow Y$ be a bijection. Prove that $f^{-1} : Y \rightarrow X$ is a bijection.

<

 Exercise 3.2.20

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f : X \rightarrow Z$ is a bijection, and write an expression for its inverse in terms of f^{-1} and g^{-1} .

<

 Exercise 3.2.21

Let $f : X \rightarrow A$ and $g : Y \rightarrow B$ be bijections. Prove that there is a bijection $X \times Y \rightarrow A \times B$, and describe its inverse.

<

Section 3.E

Chapter 3 exercises

3.E.1. For each of the following equations, determine whether there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$, the equation holds if and only if $y = f(x)$.

(a) $x + y = 1$

(c) $x = 0$

(e) $(1 + x^2)y = 1$

(b) $x^2 + y^2 = 1$

(d) $y = 0$

(f) $(1 - x^2)y = 0$

3.E.2. Let X be a set. Prove that

$$\forall a \in X, \exists ! U \in \mathcal{P}(X), (a \in U \wedge \exists ! x \in X, x \in U)$$

Give an explicit description of the function $X \rightarrow \mathcal{P}(X)$ that is suggested by this logical formula.

3.E.3. Show that there is only one function whose codomain is empty. What is its domain?

◆ **Definition 3.E.1**

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **even** if $f(-x) = f(x)$ for all $x \in \mathbb{R}$, and it is **odd** if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$.

3.E.4. Let $n \in \mathbb{N}$. Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^n$ for all $x \in \mathbb{R}$ is even if and only if n is even, and odd if and only if n is odd.

3.E.5. Prove that there is a unique function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is both even and odd.

3.E.6. Let $U \subseteq \mathbb{R}$, and let $\chi_U : \mathbb{R} \rightarrow [2]$ be the indicator function of U .

(a) Prove that χ_U is an even function if and only if $U = \{-u \mid u \in U\}$;

(b) Prove that χ_U is an odd function if and only if $\mathbb{R} \setminus U = \{-u \mid u \in U\}$.

3.E.7. Prove that for every function $f : \mathbb{R} \rightarrow \mathbb{R}$, there is a unique even function $g : \mathbb{R} \rightarrow \mathbb{R}$ and a unique odd function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = g(x) + h(x)$ for all $x \in \mathbb{R}$.

3.E.8. Let $\{\theta_n : [n] \rightarrow [n] \mid n \in \mathbb{N}\}$ be a family of functions such that $f \circ \theta_m = \theta_n \circ f$ for all $f : [m] \rightarrow [n]$. Prove that $\theta_n = \text{id}_{[n]}$ for all $n \in \mathbb{N}$.

3.E.9. Let X be a set and let $U, V \subseteq X$. Describe the indicator function $\chi_{U \Delta V}$ of the symmetric difference of U and V (see [Definition 2.E.1](#)) in terms of χ_U and χ_V .

3.E.10. [This question assumes some familiarity with integral calculus.] A lie that is commonly told to calculus students is that

$$\int \frac{1}{x} dx = \log(|x|) + c$$

where $\log$ denotes the natural logarithm function and c is an arbitrary real constant. Prove that it is actually the case that

$$\int \frac{1}{x} dx = \log(|x|) + a\chi_U(x) + b\chi_V(x)$$

where a and b are arbitrary real constants, and U and V are particular (inhabited) subsets of $\mathbb{R} \setminus \{0\}$ that you should determine.

Images and preimages

In Questions 3.E.11 to 3.E.14, find the image $f[U]$ of the subset U of the domain of the function f described in the question.

3.E.11. $f : \mathbb{R} \rightarrow \mathbb{R}; f(x) = \sqrt{1+x^2}$ for all $x \in \mathbb{R}; U = \mathbb{R}$.

3.E.12. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}; f(a, b) = a + 2b$ for all $(a, b) \in \mathbb{Z} \times \mathbb{Z}; U = \{1\} \times \mathbb{Z}$.

3.E.13. $f : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N}); f(0) = \emptyset$ and $f(n+1) = f(n) \cup \{n\}$ for all $n \in \mathbb{N}; U = \mathbb{N}$.

3.E.14. $f : \mathbb{R}^{\mathbb{R}} \rightarrow \mathbb{R}^{\mathbb{R}}$ (where $\mathbb{R}^{\mathbb{R}}$ is the set of all functions $\mathbb{R} \rightarrow \mathbb{R}$); $f(h)(x) = h(|x|)$ for all $h \in \mathbb{R}^{\mathbb{R}}$ and all $x \in \mathbb{R}; U = \mathbb{R}^{\mathbb{R}}$.

In Questions 3.E.15 to 3.E.17, find the preimage $f^{-1}[V]$ of the subset V of the codomain of the function f described in the question.

3.E.15. $f : \mathbb{R} \rightarrow \mathbb{R}; f(x) = \sqrt{1+x^2}$ for all $x \in \mathbb{R}; V = (-5, 5]$.

3.E.16. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}; f(a, b) = a + 2b$ for all $(a, b) \in \mathbb{Z} \times \mathbb{Z}; V = \{n \in \mathbb{Z} \mid n \text{ is odd}\}$.

3.E.17. $f : \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N}); f(A, B) = A \cap B$ for all $(A, B) \in \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}); V = \{\emptyset\}$.

3.E.18. Let $f : X \rightarrow Y$ be a function. For each of the following statements, either prove it is true or find a counterexample.

- (a) $U \subseteq f^{-1}[f[U]]$ for all $U \subseteq X$;
- (b) $f^{-1}[f[U]] \subseteq U$ for all $U \subseteq X$;
- (c) $V \subseteq f[f^{-1}[V]]$ for all $V \subseteq Y$;
- (d) $f[f^{-1}[V]] \subseteq V$ for all $V \subseteq Y$.

3.E.19. Let $f : X \rightarrow Y$ be a function, let A be a set, and let $p : X \rightarrow A$ and $i : A \rightarrow Y$ be functions such that the following conditions hold:

- (i) i is injective;
- (ii) $i \circ p = f$; and
- (iii) If $q : X \rightarrow B$ and $j : B \rightarrow Y$ are functions such that j is injective and $j \circ q = f$, then there is a unique function $u : A \rightarrow B$ such that $j \circ u = i$.

Prove that there is a unique bijection $v : A \rightarrow f[X]$ such that $i(a) = v(a)$ for all $a \in f[X]$.

3.E.20. Let $f : X \rightarrow Y$ be a function and let $U, V \subseteq Y$. Prove that:

- (a) $f^{-1}[U \cap V] = f^{-1}[U] \cap f^{-1}[V]$;
- (b) $f^{-1}[U \cup V] = f^{-1}[U] \cup f^{-1}[V]$; and
- (c) $f^{-1}[Y \setminus U] = X \setminus f^{-1}[U]$.

Thus preimages preserve the basic set operations.

3.E.21. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions.

- (a) Prove that $(g \circ f)[U] = g[f[U]]$ for all $U \subseteq X$;
- (b) Prove that $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$ for all $W \subseteq Z$.

3.E.22. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. Define

$$S = \{(x, y, z) \in X \times Y \times Z \mid y = f(x) \text{ and } z = g(y)\}$$

and let $p : S \rightarrow X \times Z$ be the function defined by $p(x, y, z) = (x, z)$ for all $(x, y, z) \in S$. Prove that $p[S] = \text{Gr}(g \circ f)$.

Injections, surjections and bijections

3.E.23. (a) Prove that, for all functions $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, if $g \circ f$ is bijective, then f is injective and g is surjective.

(b) Find an example of a function $f : X \rightarrow Y$ and a function $g : Y \rightarrow Z$ such that $g \circ f$ is bijective, f is not surjective and g is not injective.

3.E.24. For each of the following pairs (U, V) of subsets of $\mathbb{R}$, determine whether the specification ‘ $f(x) = x^2 - 4x + 7$ for all $x \in U$ ’ defines a function $f : U \rightarrow V$ and, if it does, determine whether f is injective and whether f is surjective.

(a) $U = \mathbb{R}$ and $V = \mathbb{R}$;

(d) $U = (3, 4]$ and $V = [4, 7)$;

(b) $U = (1, 4)$ and $V = [3, 7)$;

(e) $U = [2, \infty)$ and $V = [3, \infty)$;

(c) $U = [3, 4)$ and $V = [4, 7)$;

(f) $U = [2, \infty)$ and $V = \mathbb{R}$.

3.E.25. For each of the following pairs of sets X and Y , find (with proof) a bijection $f : X \rightarrow Y$.

(a) $X = \mathbb{Z}$ and $Y = \mathbb{N}$;

(b) $X = \mathbb{R}$ and $Y = (-1, 1)$;

(c) $X = [0, 1]$ and $Y = (0, 1)$;

(d) $X = [a, b]$ and $Y = (c, d)$, where $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$.

3.E.26. Prove that the function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(a, b) = \binom{a+b+1}{2} + b$ for all $(a, b) \in \mathbb{N} \times \mathbb{N}$ is a bijection.

3.E.27. Let $e : X \rightarrow X$ be a function such that $e \circ e = e$. Prove that there exist a set Y and functions $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $g \circ f = e$ and $f \circ g = \text{id}_Y$.

True–False questions

In Questions 3.E.28 to 3.E.35, determine (with proof) whether the statement is true or false.

3.E.28. The set $G = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 = y^2\}$ is the graph of a function.

3.E.29. The set $G = \{(x, y) \in \mathbb{Z} \times \mathbb{N} \mid x^2 = y^2\}$ is the graph of a function.

3.E.30. Every function with empty domain has an empty codomain.

3.E.31. Every function with empty codomain has an empty domain.

3.E.32. The image of a function is a subset of its domain.

3.E.33. Given a function $f : X \rightarrow Y$, the assignment $V \mapsto f^{-1}[V]$ defines a function $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$.

3.E.34. Let f , g and h be composable functions. If $h \circ g \circ f$ is injective, then g is injective.

3.E.35. Every left inverse is surjective and every right inverse is injective.

Always–Sometimes–Never questions

In [Questions 3.E.36 to 3.E.47](#), determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

3.E.36. Let $f : X \rightarrow Y$ be a function and let $U \subseteq X$. Then there is a function $g : U \rightarrow Y$ defined by $g(x) = f(x)$ for all $x \in U$.

3.E.37. Let $f : X \rightarrow Y$ be a function and let $V \subseteq Y$. Then there is a function $g : X \rightarrow V$ defined by $g(x) = f(x)$ for all $x \in X$.

3.E.38. Let X be a set. Then there is a unique function $X \rightarrow \{0\}$.

3.E.39. Let X be a set. Then there is a unique function $X \rightarrow \emptyset$.

3.E.40. Let X and Y be sets and let $G \subseteq X \times Y$. Then G is the graph of a function $f : X \rightarrow Y$.

3.E.41. Let $f : \{1, 2, 3\} \rightarrow \{1, 2\}$, let $G = \text{Gr}(f)$ and let $G' = \{(y, x) \mid (x, y) \in G\}$. Then G' is the graph of a function $\{1, 2\} \rightarrow \{1, 2, 3\}$.

3.E.42. Let $f : X \rightarrow Y$ be a function and let $U \subseteq X$ be inhabited. Then $f[U]$ is inhabited.

3.E.43. Let $f : X \rightarrow Y$ be a function and let $V \subseteq Y$ be inhabited. Then $f^{-1}[V]$ is inhabited.

3.E.44. Let $f : \{1, 2\} \rightarrow \{1, 2, 3\}$ be a function. Then f is injective.

3.E.45. Let $f : \{1, 2\} \rightarrow \{1, 2, 3\}$ be a function. Then f is surjective.

3.E.46. Let $a, b, c, d \in \mathbb{R}$ and define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $f(x, y) = (ax + by, cx + dy)$ for all $(x, y) \in \mathbb{R}^2$. Then f is injective if and only if f is surjective.

3.E.47. Let $U, V \subseteq \mathbb{R}$ and suppose that $x^2 \in V$ for all $x \in U$. The function $f : U \rightarrow V$ defined by $f(x) = x^2$ is injective.

Chapter 4

Mathematical induction

Section 4.1

Peano's axioms

 Exercise 4.1.1

Given $m \in \mathbb{N}$, define m^n for all $n \in \mathbb{N}$ by recursion on n , and prove that $2^2 = 4$ using the recursive definitions of exponentiation, multiplication and addition. ◁

 Exercise 4.1.2

Let $x_1, x_2 \in \mathbb{R}$. Working strictly from the definitions of indexed sum and indexed product, prove that

$$\sum_{i=1}^2 x_i = x_1 + x_2 \quad \text{and} \quad \prod_{i=1}^2 x_i = x_1 \cdot x_2$$
◁

 Exercise 4.1.3

Write down the next two rows of Pascal's triangle. ◁

Section 4.2

Weak induction

 Exercise 4.2.1

Prove by induction that $\sum_{k=0}^n 2^k = 2^{n+1} - 1$ for all $n \in \mathbb{N}$.

◁

 Exercise 4.2.2

Write down the statement $p(n)$ that ?? attempted to prove for all $n \geq 1$. Convince yourself that the proof of the base case is correct, then write down—with quantifiers—exactly the proposition that the induction step is meant to prove. Explain why the argument in the induction step failed to prove this proposition.

◁

 Exercise 4.2.3

Let $a, r \in \mathbb{R}$ with $r \neq 1$. Then

$$\sum_{k=0}^n ar^k = \frac{a(1 - r^{n+1})}{1 - r}$$

for all $n \in \mathbb{N}$.

◁

 Exercise 4.2.4

Let $n \in \mathbb{N}$ and for each $k \in [n+1]$ let X_k be a set. Prove by induction on n that there is a bijection

$$\prod_{k \in [n+1]} X_k \rightarrow \left(\prod_{k \in [n]} X_k \right) \times X_n.$$

◁

 Exercise 4.2.5

Prove that $\binom{n}{k} = \binom{n}{n-k}$ for all $n, k \in \mathbb{N}$ with $k \leq n$.

◁

 Exercise 4.2.6

Use the binomial theorem to prove that

$$\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$$

◁

Section 4.3

Strong induction **Exercise 4.3.1**

Define a sequence recursively by $a_0 = 4$, $a_1 = 9$ and $a_n = 5a_{n-1} - 6a_{n-2}$ for all $n \geq 2$. Prove that $a_n = 3 \cdot 2^n + 3^n$ for all $n \in \mathbb{N}$. ◁

 Exercise 4.3.2

The *Tribonacci sequence* is the sequence $t_0, t_1, t_2, \dots$ defined by

$$t_0 = 0, \quad t_1 = 0, \quad t_2 = 1, \quad t_n = t_{n-1} + t_{n-2} + t_{n-3} \text{ for all } n \geq 3$$

Prove that $t_n \leq 2^{n-3}$ for all $n \geq 3$. ◁

 Exercise 4.3.3

The *Frobenius coin problem* asks when a given amount of money can be obtained from coins of given denominations. For example, a value of 7 dubloons cannot be obtained using only 3 dubloon and 5 dubloon coins, but for all $n \geq 8$, a value of n dubloons *can* be obtained using only 3 dubloon and 5 dubloon coins. Prove this. ◁

 Exercise 4.3.4

What goes wrong in the proof of ?? if we try instead to prove that $\sqrt{4}$ is irrational? ◁

 Exercise 4.3.5

Prove that $\sqrt{3}$ is irrational. ◁

 Exercise 4.3.6

Prove that the interval $(0, 1)$ has infinite descent. ◁

 Exercise 4.3.7

Let $n \in \mathbb{N}$. By considering the expression $\sqrt{n} \times \frac{\sqrt{n} - \lfloor \sqrt{n} \rfloor}{\sqrt{n} - \lfloor \sqrt{n} \rfloor}$, prove by infinite descent that if $\sqrt{n}$ is not an integer, then $\sqrt{n}$ is irrational. ◁

Section 4.E

Chapter 4 exercises

Recursive definitions

In Questions 4.E.1 to 4.E.5, use the recursive definitions of addition, multiplication and exponentiation directly to prove the desired equation.

4.E.1. $1 + 3 = 4$

4.E.2. $0 + 5 = 5$

4.E.3. $2 \cdot 3 = 6$

4.E.4. $0 \cdot 5 = 0$

4.E.5. $2^3 = 8$

4.E.6. Give a recursive definition of new quantifiers $\exists^n$ for $n \in \mathbb{N}$, where given a set X and a predicate $p(x)$, the logical formula $\exists^n x \in X, p(x)$ means ‘there are exactly n elements $x \in X$ such that $p(x)$ is true’. That is, define $\exists^0$, and then define $\exists^{n+1}$ in terms of $\exists^n$.

4.E.7. Use the recursive definition of binomial coefficients (??) to prove directly that $\binom{4}{2} = 6$.

4.E.8. (a) Find the number of trailing 0s in the decimal expansion of $41!$.

(b) Find the number of trailing 0s in the binary expansion of $41!$.

4.E.9. Let N be a set, let $z \in N$ and let $s : N \rightarrow N$. Prove that (N, z, s) is a notion of natural numbers (in the sense of ??) if and only if, for every set X , every element $a \in X$ and every function $f : X \rightarrow X$, there is a unique function $h : N \rightarrow X$ such that $h(z) = a$ and $h \circ f = s \circ h$.

Proofs by induction

4.E.10. Find all natural numbers n such that $n^5 < 5^n$.

4.E.11. Prove that $(1+x)^{123\,456\,789} \geq 1 + 123\,456\,789x$ for all real $x \geq -1$.

4.E.12. Let $a \in \mathbb{N}$ and assume that the last digit in the decimal expansion of a is 6. Prove that the last digit in the decimal expansion of a^n is 6 for all $n \geq 1$.

4.E.13. Let $a, b \in \mathbb{R}$, and let $a_0, a_1, a_2, \dots$ be a sequence such that $a_0 = a$, $a_1 = b$ and $a_n = \frac{a_{n-1} + a_{n+1}}{2}$ for all $n \geq 1$. Prove that $a_n = a + (b-a)n$ for all $n \in \mathbb{N}$.

4.E.14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

(a) Prove by induction that there is a real number a such that $f(n) = an$ for all $n \in \mathbb{N}$.

(b) Deduce that $f(n) = an$ for all $n \in \mathbb{Z}$.

(c) Deduce further that $f(x) = ax$ for all $x \in \mathbb{Q}$.

4.E.15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(0) > 0$ and $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Prove that there is some positive real number a such that $f(x) = a^x$ for all *rational* numbers x .

4.E.16. Let $a, b \in \mathbb{Z}$. Prove that $b - a$ divides $b^n - a^n$ for all $n \in \mathbb{N}$.

4.E.17. Prove by induction that $7^n - 2 \cdot 4^n + 1$ is divisible by 18 for all $n \in \mathbb{N}$.

Some examples from calculus

Questions 4.E.18 to 4.E.23 assume familiarity with the basic techniques of differential and integral calculus.

4.E.18. Prove that $\frac{d^n}{dx^n}(xe^x) = (n+x)e^x$ for all $n \in \mathbb{N}$.

4.E.19. Find and prove an expression for $\frac{d^n}{dx^n}(x^2e^x)$ for all $n \in \mathbb{N}$.

4.E.20. Let f and g be differentiable functions. Prove that

$$\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x)g^{(n-k)}(x)$$

for all $n \in \mathbb{N}$, where the notation $h^{(r)}$ denotes the r^{th} derivative of h .

4.E.21. Find and prove an expression for the n^{th} derivative of $\log(x)$ for all $n \in \mathbb{N}$.

4.E.22. Prove that $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ for all $\theta \in \mathbb{R}$ and all $n \in \mathbb{N}$.

4.E.23. (a) Prove that $\int_0^{\frac{\pi}{2}} \sin^{n+2}(x) dx = \frac{n+1}{n+2} \int_0^{\frac{\pi}{2}} \sin^n(x) dx$ for all $n \in \mathbb{N}$.

(b) Use induction to prove that $\int_0^{\frac{\pi}{2}} \sin^{2n}(x) dx = \frac{\pi}{2^{2n+1}} \binom{2n}{n}$ for all $n \in \mathbb{N}$.

(c) Find and prove an expression for $\int_0^{\frac{\pi}{2}} \sin^{2n+1}(x) dx$ for all $n \in \mathbb{N}$.

True–False questions

In **Questions 4.E.24 to 4.E.32**, determine (with proof) whether the statement is true or false.

4.E.24. Every factorial is positive.

4.E.25. Every binomial coefficient is positive.

4.E.26. Every proof by strong induction can be turned into a proof by weak induction by changing only the induction hypothesis.

4.E.27. Every proof by weak induction can be turned into a proof by strong induction by changing only the induction hypothesis.

4.E.28. Given a set X , a surjection $f : \mathbb{N} \rightarrow X$ and a logical formula $p(x)$ with free variable $x \in X$, if $p(f(0))$ is true and $\forall n \in \mathbb{N}, (p(f(n)) \Rightarrow p(f(n+1)))$ is true, then $\forall x \in X, p(x)$ is true.

4.E.29. Given some logical formula $p(x)$ with free variable $x \in \mathbb{N}$, if for all $x \in \mathbb{N}$ there exists some $y \leq x$ such that $p(y)$ is false, then $p(x)$ is false for all $x \in \mathbb{N}$.

4.E.30. Every inhabited subset of the set $\mathbb{Q}^{\geq 0}$ of nonnegative rational numbers has a least element.

4.E.31. Every inhabited subset of the set $\mathbb{Z}$ of integers has a least element.

4.E.32. Every inhabited subset of the set $\mathbb{Z}^-$ of negative integers has a greatest element.

Always–Sometimes–Never questions

In [Questions 4.E.33 to 4.E.35](#), determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

4.E.33. Let $n, k, \ell \in \mathbb{Z}$. Then $\binom{n}{k+\ell} = \binom{n}{k} + \binom{n}{\ell}$.

4.E.34. Let $m, n \geq 2$. Then $(m+n)! = m! + n!$.

4.E.35. Let $p(x)$ be a logical formula with free variable $x \in \mathbb{Z}$, and suppose that $p(0)$ is true and, for all $n \in \mathbb{Z}$, if $p(n)$ is true, then $p(n+1)$ and $p(n-2)$ are true. Then $\forall n \in \mathbb{Z}, p(n)$ is true.

Chapter 5

Relations

Section 5.1

Relations

 Exercise 5.1.1

Give three more examples of relations, not all of which are homogeneous. ◁

 Exercise 5.1.2

Let R and S be relations on $\mathbb{R}$ defined for $a, b \in \mathbb{R}$ by letting

$$a R b \Leftrightarrow b - a \in \mathbb{Q} \quad \text{and} \quad a S b \Leftrightarrow \exists n \in \mathbb{Z}, (n \neq 0) \wedge n(b - a) \in \mathbb{Z}$$

Prove that $R = S$. ◁

 Exercise 5.1.3

Let R be the relation on $\mathbb{Z}$ defined for $x, y \in \mathbb{Z}$ by letting $x R y$ if and only if $x^2 = y^2$. Describe its graph $\text{Gr}(R) \subseteq \mathbb{Z} \times \mathbb{Z}$. ◁

 Exercise 5.1.4

Let $f : X \rightarrow Y$ be a function, and define the relation R_f from X to Y as in ???. Prove that $\text{Gr}(R_f) = \text{Gr}(f)$ —that is, the graph of the *relation* R_f is equal to the graph of the *function* f . ◁

 Exercise 5.1.5

Let X and Y be sets. Describe the graphs $\text{Gr}(D_{X,Y})$ and $\text{Gr}(\emptyset_{X,Y})$. ◁

 Exercise 5.1.6

Let X be a set. Prove that $\Delta_X = \text{Gr}(=)$. ◁

 Exercise 5.1.7

Let X be a set. Prove that $\subseteq$ is a reflexive relation on $\mathcal{P}(X)$, but $\subsetneq$ is not. ◁

 Exercise 5.1.8

Prove that the relation ‘ x divides y ’ on $\mathbb{Z}$ is reflexive. ◁

 Exercise 5.1.9

Let $G = \{(0, 0), (1, 1), (2, 2)\}$. Let R be the relation on $[3]$ whose graph is G , and let S be the relation on $[4]$ whose graph is G . Prove that R is reflexive, but S is not. ◁

 Exercise 5.1.10

Find all subsets $U \subseteq \mathbb{Z}$ such that the relation ‘ x divides y ’ on U is symmetric. ◁

 Exercise 5.1.11

Let R and S be relations such that $\text{Gr}(R) = \text{Gr}(S)$. Note that the domain of R might be different from the domain of S . Prove that R is symmetric if and only if S is symmetric. ◁

 Exercise 5.1.12

Prove that the relation ‘ x divides y ’ on $\mathbb{N}$ is antisymmetric, but not on $\mathbb{Z}$. ◁

 Exercise 5.1.13

Let X be a set. Prove that $\subseteq$ is an antisymmetric relation on $\mathcal{P}(X)$. ◁

 Exercise 5.1.14

Let X be a set and let R be a relation on X . Prove that R is both symmetric and antisymmetric if and only if $\text{Gr}(R) \subseteq \Delta_X$, where Δ_X is the diagonal subset of $X \times X$ (see ??). Deduce that the only reflexive, symmetric and antisymmetric relation on a set X is the equality relation on X . ◁

 Exercise 5.1.15

Let X be a set. Prove that $\subseteq$ is a transitive relation on $\mathcal{P}(X)$. ◁

 Exercise 5.1.16

Prove that the relation ‘ x divides y ’ on $\mathbb{Z}$ is transitive. ◁

 Exercise 5.1.17

Let R and S be relations such that $\text{Gr}(R) = \text{Gr}(S)$. Note that the domain of R might be different from the domain of S . Prove that R is transitive if and only if S is transitive. ◁

Section 5.2

Equivalence relations and partitions

 Exercise 5.2.1

Given a function $f : X \rightarrow Y$, define a relation $\sim_f$ on X by

$$a \sim_f b \iff f(a) = f(b)$$

for all $a, b \in X$. Prove that $\sim_f$ is an equivalence relation on X . ◁

 Exercise 5.2.2

Let $\mathcal{S}$ be some set whose elements are all sets. (For example, we could take $\mathcal{S} = \mathcal{P}(X)$ for some fixed set X .) Define a relation $\cong$ ([L^AT_EX code: \cong](#)) on $\mathcal{S}$ by letting $U \cong V$ if and only if there exists a bijection $f : U \rightarrow V$, for all $U, V \in \mathcal{S}$. Prove that $\cong$ is an equivalence relation on $\mathcal{S}$. ◁

 Exercise 5.2.3

Let $n \in \mathbb{Z}$. Prove that if $n \neq 0$, then $a \equiv b \pmod{n}$ if and only if a and b have the same remainder when divided by n . ◁

 Exercise 5.2.4

Let $a, b \in \mathbb{Z}$. When is it true that $a \equiv b \pmod{0}$? When is it true that $a \equiv b \pmod{1}$? ◁

 Exercise 5.2.5

Let $\approx$ be the relation of congruence modulo 10 on $\mathbb{N}$. Describe the equivalence classes, and give an explicit expression of the quotient $\mathbb{N}/\approx$ in list notation. ◁

 Exercise 5.2.6

Let $f : X \rightarrow Y$ be a function. Prove that f is injective if and only if each $\sim_f$ -equivalence class has a unique element, where $\sim_f$ is the equivalence relation defined in [Exercise 5.2.1](#). ◁

 Exercise 5.2.7

Let $n \in \mathbb{Z}$. Prove that the function

$$i : [|n|] \rightarrow \mathbb{Z}/n\mathbb{Z}$$

defined by $i(r) = [r]_n$ for all $r \in [|n|]$ is a bijection. ◁

 Exercise 5.2.8

Let $f : X \rightarrow Y$ be a surjection, and define a collection $\mathcal{F}$ of subsets of X by

$$\mathcal{F} = \{f^{-1}[\{b\}] \mid b \in Y\}$$

That is, $\mathcal{F}$ is the set of subsets of X given by the preimages of individual elements of Y under f . Prove that $\mathcal{F}$ is a partition of X . Where in your proof do you use surjectivity of f ? ◁

 Exercise 5.2.9

Let X be a set and let $\mathcal{A}$ be a set of inhabited subsets of X . Prove that $\mathcal{A}$ is a partition of X if and only if for each $x \in X$, there is a unique set $A \in \mathcal{A}$ with $x \in A$. ◁

 Exercise 5.2.10

Prove part (a) of ?? ◁

 Exercise 5.2.11

For each $i \in [0, 1)$ let $A_i = \{i + n \mid n \in \mathbb{Z}\} \subseteq \mathbb{R}$, and define $\mathcal{A} = \{A_i \mid i \in [0, 1)\}$.

Prove that $\mathcal{A}$ is a partition of $\mathbb{R}$, and provide an explicit description of the (unique!) equivalence relation $\sim$ on $\mathbb{R}$ such that $\mathbb{R}/\sim = \mathcal{A}$. ◁

 Exercise 5.2.12

Let X and Y be sets, let $\sim$ be an equivalence relation on X and let $\approx$ be an equivalence relation on Y . Assume that there is a bijection $p : X/\sim \rightarrow Y/\approx$, and for each equivalence class $E \in X/\sim$ there is a bijection $h_E : E \rightarrow p(E)$. Use ?? to prove that there is a bijection $h : X \rightarrow Y$. $\triangleleft$

 Exercise 5.2.13

Let $n \in \mathbb{Z}$ with $n \neq 0$. Describe the quotient function $q_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ in terms of remainders. $\triangleleft$

 Exercise 5.2.14

Let $\sim$ be an equivalence relation on a set X . Prove that the quotient function $q_\sim : X \rightarrow X/\sim$ is surjective. $\triangleleft$

 Exercise 5.2.15

Give an explicit description of the dashed arrow in the above diagram. That is, describe the correspondence between partitions of a set X and surjections whose domain is X . $\triangleleft$

Section 5.E

Chapter 5 exercises

Properties of relations

5.E.1. For each of the eight subsets

$$P \subseteq \{\text{reflexive, symmetric, transitive}\}$$

find a relation satisfying (only) the properties in P .

5.E.2. Prove that if R is a symmetric, antisymmetric relation on a set X , then it is a subrelation of the equality relation—that is, $\text{Gr}(R) \subseteq \text{Gr}(=)$.

5.E.3. A relation R on a set X is **left-total** if for all $x \in X$, there exists some $y \in X$ such that xRy . Prove that every left-total, symmetric, transitive relation is reflexive.

Equivalence relations

◆ Definition 5.E.1

Let R be a relation on a set X and let $f : X \rightarrow Y$ be a function. The **transport** of R along f is the relation S on Y defined for $c, d \in Y$ by letting $c S d$ if and only if there exist $a, b \in X$ such that $f(a) = c$, $f(b) = d$ and $a R b$. That is

$$\text{Gr}(S) = \{(f(a), f(b)) \mid a, b \in X, a R b\}$$

5.E.4. Let X and Y be sets and let $f : X \rightarrow Y$. Prove that if $\sim$ is an equivalence relation on X , then the transport of $\sim$ along f is an equivalence relation on Y .

◆ Definition 5.E.2

Let R be any relation on a set X . The **equivalence relation generated by R** is the relation $\sim_R$ on X defined as follows. Given $x, y \in X$, say $x \sim_R y$ if and only if for some $k \in \mathbb{N}$ there is a sequence $(a_0, a_1, \dots, a_k)$ of elements of X such that $a_0 = x$, $a_k = y$ and, for all $i \in [k]$, either $a_i R a_{i+1}$ or $a_{i+1} R a_i$.

5.E.5. Fix $n \in \mathbb{Z}$ and let R be the relation on $\mathbb{Z}$ defined by xRy if and only if $y = x + n$. Prove that $\sim_R$ is the relation of congruence modulo n .

5.E.6. Let X be a set and let R be the subset relation on $\mathcal{P}(X)$. Prove that $U \sim_R V$ for all $U, V \subseteq X$.

5.E.7. Let X be a set, fix two distinct elements $a, b \in X$, and define a relation R on X by declaring aRb only—that is, for all $x, y \in X$, we have xRy if and only if $x = a$ and $y = b$. Prove that the relation $\sim_R$ is defined by $x \sim_R y$ if and only if either $x = y$ or $\{x, y\} = \{a, b\}$.

In Questions 5.E.8 to 5.E.11, let R be a relation on a set X , and let $\sim_R$ be the equivalence relation generated by R (as in Definition 5.E.2). In these questions, you will prove that $\sim_R$ is the ‘smallest’ equivalence relation extending R .

5.E.8. Prove that $\sim_R$ is an equivalence relation on X .

5.E.9. Prove that $xRy \Rightarrow x \sim_R y$ for all $x, y \in X$.

5.E.10. Prove that if $\approx$ is any equivalence relation on X and $xRy \Rightarrow x \approx y$ for all $x, y \in X$, then $x \sim_R y \Rightarrow x \approx y$ for all $x, y \in X$.

5.E.11. Prove that if R is an equivalence relation, then $\sim_R = R$.

5.E.12. This question requires some familiarity with concepts from calculus. Let $\mathcal{C}^1(\mathbb{R})$ denote the set of all differentiable functions $F : \mathbb{R} \rightarrow \mathbb{R}$ whose derivative $F' : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, and define a relation $\sim$ on $\mathcal{C}^1(\mathbb{R})$ by letting $F \sim G$ if and only if $F' = G'$.

(a) Prove that $\sim$ is an equivalence relation on $\mathcal{C}^1(\mathbb{R})$.

(b) Let $F \in \mathcal{C}^1(\mathbb{R})$. Prove that

$$[F]_{\sim} = \{G \in \mathcal{C}^1(\mathbb{R}) \mid \exists c \in \mathbb{R}, \forall x \in \mathbb{R}, G(x) = F(x) + c\}$$

(c) Interpret the notion of an indefinite integral $\int f(x) dx$ of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ in terms of $\sim$ -equivalence classes.

True–False questions

In [Questions 5.E.13 to 5.E.19](#), determine (with proof) whether the statement is true or false.

5.E.13. Every reflexive relation is symmetric.

5.E.14. There is a unique relation on $\mathbb{N}$ that is reflexive, symmetric and antisymmetric.

5.E.15. Every relation that is not symmetric is antisymmetric.

5.E.16. Every symmetric, antisymmetric relation is reflexive.

5.E.17. The empty set is the graph of a relation on $\mathbb{N}$.

5.E.18. For all sets X , the graph of a function $X \rightarrow X$ is the graph of a relation on X .

5.E.19. For all sets X , the graph of a relation on X is the graph of a function $X \rightarrow X$.

Always–Sometimes–Never questions

In [Questions 5.E.20 to 5.E.21](#), determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

5.E.20. Let $\sim$ be an equivalence relation on a set X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$, defined by $A \approx B$ if and only if $x \sim y$ for all $x \in A$ and $y \in B$, is an equivalence relation.

5.E.21. Let $\sim$ be an equivalence relation on a set X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$, defined by $A \approx B$ if and only if $x \sim y$ for some $x \in A$ and $y \in B$, is an equivalence relation.

Chapter 6

Finite and infinite sets

Section 6.1

Finite sets

 Exercise 6.1.1

Prove that $[n]$ is finite for each $n \in \mathbb{N}$.

◁

✚ **Solution**

Let $n \in \mathbb{N}$. The identity function $\text{id}_{[n]} : [n] \rightarrow [n]$ is a bijection by ??, so is an enumeration of $[n]$.

 Exercise 6.1.2

Prove parts (b) and (c) of ??.

◁

 Exercise 6.1.3

Let X be a finite set.

(a) Prove that for all $a \in X$, the set $X \setminus \{a\}$ is finite and $|X \setminus \{a\}| = |X| - 1$; and

(b) Prove that for all $b \notin X$, the set $X \cup \{b\}$ is finite and $|X \cup \{b\}| = |X| + 1$.

What happens in part (a) if we do not require $a \in X$? What happens in part (b) if we do not require $b \notin X$?

◁

 Exercise 6.1.4

Prove parts (b) and (c) of ??.

◁

 Exercise 6.1.5

Let X be a finite set. Prove that every subset $A \subseteq X$ is finite and $|A| \leq |X|$.

◁

✚ **Solution**

Let $i : A \rightarrow X$ be the *inclusion function*, defined by $i(x) = x$ for all $x \in A$. For all $a, b \in A$, if $i(a) = i(b)$, then $a = b$ by definition of i , so that i is injective. Since X is finite and $i : A \rightarrow X$ is an injection, it follows from ????.

 Exercise 6.1.6

Let X and Y be finite sets. Prove that $X \cap Y$ is finite. ◁

 Exercise 6.1.7

The following steps complete the proof of ??:

- (a) Given sets A and B , prove that the sets $A \times \{0\}$ and $B \times \{1\}$ are disjoint, and find bijections $A \rightarrow A \times \{0\}$ and $B \rightarrow B \times \{1\}$. Write $A \sqcup B$ ([L^AT_EX code: \sqcup](#)) to denote the set $(A \times \{0\}) \cup (B \times \{1\})$. The set $A \sqcup B$ is called the **disjoint union** of A and B .

- (b) Prove that, if A and B are finite then $A \sqcup B$ is finite and

$$|A \sqcup B| = |A| + |B|$$

- (c) Let X and Y be sets. Find a bijection

$$(X \cup Y) \sqcup (X \cap Y) \rightarrow X \sqcup Y$$

- (d) Complete the proof of ??—that is, prove that if X and Y are finite sets, not necessarily disjoint, then $X \cup Y$ is finite and

$$|X \cup Y| = |X| + |Y| - |X \cap Y|$$
◁

 Exercise 6.1.8

Let X be a finite set and let $U \subseteq X$. Prove that $X \setminus U$ is finite, and moreover $|X \setminus U| = |X| - |U|$. How does this result change if U is not required to be a subset of X ? ◁

 Exercise 6.1.9

Let $m, n \in \mathbb{N}$. Prove that $|[m] \times [n]| = mn$. ◁

Section 6.2

Countable and uncountable sets

 Exercise 6.2.1

Prove that the set of all positive integers is countably infinite. ◁

 Exercise 6.2.2

Prove that the set of all even natural numbers is countably infinite, and that the set of all odd natural numbers is countably infinite. ◁

 Exercise 6.2.3

Prove that the function $p : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ defined by $p(x, y) = 2^x(2y + 1) - 1$ is a bijection. Deduce that $\mathbb{N} \times \mathbb{N}$ is countable. ◁

 Exercise 6.2.4

Prove if X and Y are countably infinite sets, then $X \times Y$ is countably infinite. ◁

 Exercise 6.2.5

Let X be a countably infinite set. In the proof of (iii) $\Rightarrow$ (i) in ??, we gave an explicit construction of a bijection $f : \mathbb{N} \rightarrow X$ using a surjection $h : \mathbb{N} \rightarrow X$. Prove (ii) $\Rightarrow$ (i) the same way—that is, explain how to construct a bijection $f : \mathbb{N} \rightarrow X$ using an injection $g : X \rightarrow \mathbb{N}$. ◁

 Exercise 6.2.6

Let X be an inhabited set. The following are equivalent:

- (i) X is countable;
- (ii) There exists an injection $g : X \rightarrow C$ for some countable set C ;
- (iii) There exists a surjection $h : C \rightarrow X$ for some countable set C .

◁

 Exercise 6.2.7

Let X be a countable set. Prove that $\binom{X}{k}$ is countable for each $k \in \mathbb{N}$. ◁

 Exercise 6.2.8

Use Cantor's diagonal argument to prove that the set $\mathbb{N}^{\mathbb{N}}$ of all functions $\mathbb{N} \rightarrow \mathbb{N}$ is uncountable. ◁

 Exercise 6.2.9

Prove that if a set X has an uncountable subset, then X is uncountable. ◁

 Exercise 6.2.10

Let X be a set. Prove that $\mathcal{P}(X)$ is either finite or uncountable. ◁

 Exercise 6.2.11

Let $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, \div, -\}$. Describe which words over Σ immediately represent rational numbers (in the sense of treating the digit symbols as numbers and the symbols $\div$ and $-$ as arithmetic operations). Find some examples of elements of Σ^* that do not represent rational numbers. ◁

 Exercise 6.2.12

Let Σ be an alphabet. Prove that if Σ is countable, then Σ^* is countable. ◁

 Exercise 6.2.13

Give an explicit finite description of the rational numbers over a finite alphabet. ◁

 Exercise 6.2.14

Let $\mathcal{C}(\mathbb{N})$ be the set of all *cofinite* subsets of $\mathbb{N}$; that is

$$\mathcal{C}(\mathbb{N}) = \{U \subseteq \mathbb{N} \mid \mathbb{N} \setminus U \text{ is finite}\}$$

Give an explicit finite description of the elements of $\mathcal{C}(\mathbb{N})$ over a finite alphabet. ◁

 Exercise 6.2.15

Prove that the set of all polynomials with rational coefficients is countable. ◁

 Exercise 6.2.16

Consider the following erroneous argument that claims to be a proof that $\mathcal{P}(\mathbb{N})$ is countable.

Let $\Sigma = \{\mathbb{N}, x, U, V, \boxed{|}, \boxed{\in}, \boxed{\{}, \boxed{\}}\}$ —again, we have boxed some symbols to emphasise that they are elements of Σ .

Define $D : \mathcal{P}(\mathbb{N}) \rightarrow \Sigma^*$ by

$$D(U) = \{x \in \mathbb{N} \mid x \in U\} \quad \text{and} \quad D(V) = \{x \in \mathbb{N} \mid x \in V\}$$

for all $U, V \subseteq \mathbb{N}$. Then Σ is finite, and D is injective since for all $U, V \subseteq \mathbb{N}$ we have

$$D(U) = D(V) \Rightarrow \{x \in \mathbb{N} \mid x \in U\} = \{x \in \mathbb{N} \mid x \in V\} \Rightarrow U = V$$

Since D is a finite description of the subsets of $\mathbb{N}$ over a finite alphabet, it follows from ?? that $\mathcal{P}(\mathbb{N})$ is countable. □

This proof cannot be valid, since we already proved in ?? that $\mathcal{P}(\mathbb{N})$ is uncountable. Where in the proof did we go wrong? ◁

 Exercise 6.2.17

Suppose we were to attempt to prove that $\mathcal{P}(\mathbb{N})$ is countable using the same argument as ??. Where does the proof fail? ◁

 Exercise 6.2.18

Prove that, for every countable set X , there is a finite description of the elements of X over a *finite* alphabet. ◁

Section 6.E

Chapter 6 exercises

Part II

Topics in pure mathematics

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Chapter 7

Number theory

Section 7.1

Division

 Exercise 7.1.1

Show that if $a \in \mathbb{Z}$ then $1 \mid a$, $-1 \mid a$ and $a \mid 0$. For which integers a does $a \mid 1$? For which integers a does $0 \mid a$? ◁

 Exercise 7.1.2

There are two greatest common divisors of 6 and 15; find both. ◁

 Exercise 7.1.3

Let $a, b \in \mathbb{Z}$. If d_1 and d_2 are two greatest common divisors of a and b , then either $d_1 = d_2$ or $d_1 = -d_2$. ◁

 Exercise 7.1.4

For each $n \in \mathbb{Z}$, let $D_n \subseteq \mathbb{Z}$ be the set of divisors of n . Prove that $D_a \cap D_b = D_{\gcd(a,b)}$ for all $a, b \in \mathbb{Z}$. ◁

 Exercise 7.1.5

Prove the following facts about greatest common divisors:

(a) $\gcd(a, b) = \gcd(|a|, |b|)$ for all $a, b \in \mathbb{Z}$.

(b) $\gcd(a, b) = \gcd(b, a)$ for all $a, b \in \mathbb{Z}$.

(c) $\gcd(a, 0) = |a|$ for all $a \in \mathbb{Z}$. ◁

 Exercise 7.1.6

Which of the following equations have solutions?

(a) $12u + 9v = -18$

(b) $12u + 9v = 1$

(c) $100u + 35v = 125$

(d) $7125u + 1300v = 0$

(e) $1010u + 101010v = 1010101010101010$

(f) $14u - 4v = 12$

◁

 Exercise 7.1.7

Which integers in the set $[15]$ are coprime to 15?

◁

 Exercise 7.1.8

Let a and b be integers, not both zero, and let $d = \gcd(a, b)$. The integers $\frac{a}{d}$ and $\frac{b}{d}$ are coprime.

◁

 Exercise 7.1.9

Find a solution $(x, y) \in \mathbb{Z} \times \mathbb{Z}$ to the equation $630x + 385y = 4340$.

◁

 Exercise 7.1.10

Find all integers x, y such that

$$630x + 385y = 4340$$

◁

 Exercise 7.1.11

Let $a, b \in \mathbb{Z}$. Prove that a and b have a least common multiple. Furthermore, prove that least common multiples are unique up to sign, in the sense that if m, m' are two least common multiples of a and b , then $m = m'$ or $m = -m'$.

◁

 Exercise 7.1.12

Let $a, b \in \mathbb{Z}$. Prove that $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$.

◁

Section 7.2

Prime numbers

 Exercise 7.2.1

Using ??, prove that 3 and 5 are prime, and that 4 is not prime. ◁

 Exercise 7.2.2

Let $p \in \mathbb{Z}$ be a positive prime and let $0 < k < p$. Show that $p \mid \binom{p}{k}$. ◁

 Exercise 7.2.3

Let $p \in \mathbb{Z}$. Prove that if p is ring theoretically prime, then p is irreducible. ◁

 Exercise 7.2.4

Express the following numbers as products of primes:

$$16 \quad -240 \quad 5050 \quad 111111 \quad -123456789$$

◁

 Exercise 7.2.5

Write out the canonical prime factorisations of the numbers from [Exercise 7.2.4](#), which were:

$$16 \quad -240 \quad 5050 \quad 111111 \quad -123456789$$

◁

 Exercise 7.2.6

Let $p_1, p_2, \dots, p_r$ be distinct primes, and let $k_i, \ell_i \in \mathbb{N}$ for all $1 \leq i \leq r$. Define

$$m = p_1^{k_1} \times p_2^{k_2} \times \cdots \times p_r^{k_r} \quad \text{and} \quad n = p_1^{\ell_1} \times p_2^{\ell_2} \times \cdots \times p_r^{\ell_r}$$

Prove that

$$\gcd(m, n) = p_1^{u_1} \times p_2^{u_2} \times \cdots \times p_r^{u_r}$$

where $u_i = \min\{k_i, \ell_i\}$ for all $1 \leq i \leq r$. ◁

 Exercise 7.2.7

Let P be an inhabited finite set of positive prime numbers and let m be the product of all the elements of P . That is, for some $n \geq 1$ let

$$P = \{p_1, \dots, p_n\} \quad \text{and} \quad m = p_1 \times \cdots \times p_n$$

where each $p_k \in P$ is a positive prime. Using the fundamental theorem of arithmetic, show that $m + 1$ has a positive prime divisor which is not an element of P . ◁

 Exercise 7.2.8

Let $n \in \mathbb{Z}$ with $n > 2$. Prove that the set $\{k \in \mathbb{Z} \mid n < k < n!\}$ contains a prime number. ◁

Section 7.3

Modular arithmetic

 Exercise 7.3.1

For which integers x does the congruence $5x + 1 \equiv x + 8 \pmod{3}$ hold? Characterise such integers x in terms of their remainder when divided by 3. ◁

 Exercise 7.3.2

Fix a modulus n . Is it true that

$$a \equiv b \pmod{n} \Rightarrow a^k \equiv b^k \pmod{n}$$

for all $a, b \in \mathbb{Z}$ and $k \in \mathbb{N}$? If so, prove it; if not, provide a counterexample. ◁

 Exercise 7.3.3

Fix a modulus n . Is it true that

$$k \equiv \ell \pmod{n} \Rightarrow a^k \equiv a^\ell \pmod{n}$$

for all $k, \ell \in \mathbb{N}$ and $a \in \mathbb{Z}$? If so, prove it; if not, provide a counterexample. ◁

 Exercise 7.3.4

Fix a modulus n . Is it true that

$$qa \equiv qb \pmod{n} \Rightarrow a \equiv b \pmod{n}$$

for all $a, b, q \in \mathbb{Z}$ with $q \not\equiv 0 \pmod{n}$? If so, prove it; if not, provide a counterexample. ◁

 Exercise 7.3.5

For $n = 7, 8, 9, 10, 11, 12$, either find a multiplicative inverse for 6 modulo n , or show that no multiplicative inverse exists. Can you spot a pattern? ◁

 Exercise 7.3.6

Let n be a modulus and let $a \in \mathbb{Z}$. Suppose that u is a multiplicative inverse for a modulo n . Prove that, for all $k \in \mathbb{Z}$, $u + kn$ is a multiplicative inverse for a modulo n . ◁

 Exercise 7.3.7

Find all integers x such that $25x - 4 \equiv 4x + 3 \pmod{13}$. ◁

 Exercise 7.3.8

Let n be a modulus and let $a \in \mathbb{Z}$ with $a \perp n$. Prove that there exists $k \geq 1$ such that $a^k \equiv 1 \pmod{n}$. ◁

 Exercise 7.3.9

Find the remainder of 3^{244886} when divided by 13. ◁

 Exercise 7.3.10

Let $n \in \mathbb{Z}$ and let $p > 0$ be prime. Prove that if $p \mid n$, then $\varphi(pn) = p \cdot \varphi(n)$. Deduce that $\varphi(p^k) = p^k - p^{k-1}$ for all prime $p > 0$ and all $k \geq 1$. ◁

 Exercise 7.3.11

Let $n \in \mathbb{Z}$ and let $p > 0$ be prime. Prove that if $p \nmid n$, then $\varphi(pn) = (p-1)\varphi(n)$. ◁

 Exercise 7.3.12

Prove that $\varphi(100) = 40$, this time using the inclusion–exclusion principle. ◁

 Exercise 7.3.13

Use Euler's theorem to prove that the last two digits of 3^{79} are '67'.

◁

 Exercise 7.3.14

Let $n \in \mathbb{Z}$ be composite. Prove that if $n > 4$, then $n \mid (n-1)!$.

◁

 Exercise 7.3.15

Let p be a positive prime and let $a \in \mathbb{Z}$. Prove that, if $a^2 \equiv 1 \pmod{p}$, then $a \equiv 1 \pmod{p}$ or $a \equiv -1 \pmod{p}$.

◁

 Exercise 7.3.16

Let p be a positive prime. Prove that $(p-1)! \equiv -1 \pmod{p}$.

◁

 Exercise 7.3.17

Let p be an odd positive prime. Prove that

$$\left[\left(\frac{p-1}{2} \right)! \right]^2 \equiv (-1)^{\frac{p+1}{2}} \pmod{p}$$

◁

 Exercise 7.3.18

Find all integer solutions x to the system of congruences:

$$\begin{cases} x \equiv -1 \pmod{4} \\ x \equiv 1 \pmod{9} \\ x \equiv 5 \pmod{11} \end{cases}$$

Express your solution in the form $x \equiv a \pmod{n}$ for suitable $n > 0$ and $0 \leq a < n$.

◁

 Exercise 7.3.19

Let m, n be coprime moduli and let $a, b \in \mathbb{Z}$. Let $u, v \in \mathbb{Z}$ be such that

$$mu \equiv 1 \pmod{n} \quad \text{and} \quad nv \equiv 1 \pmod{m}$$

In terms of a, b, m, n, u, v , find an integer x such that

$$x \equiv a \pmod{m} \quad \text{and} \quad x \equiv b \pmod{n}$$

◁

 Exercise 7.3.20

Let m, n be coprime moduli and let $x, y \in \mathbb{Z}$. Prove that if $x \equiv y \pmod{m}$ and $x \equiv y \pmod{n}$, then $x \equiv y \pmod{mn}$.

◁

 Exercise 7.3.21

Verify that the algorithm outlined above is correct. Use it to compute the solutions to the system of congruences

$$x \equiv 3 \pmod{12} \quad \text{and} \quad x \equiv 15 \pmod{20}$$

◁

 ★ Exercise 7.3.22

Generalise the Chinese remainder theorem to systems of arbitrarily (finitely) many congruences. That is, given $r \in \mathbb{N}$, find precisely the conditions on moduli $n_1, n_2, \dots, n_r$ and integers $a_1, a_2, \dots, a_r$ such that an integer solution exists to the congruences

$$x \equiv a_1 \pmod{n_1}, \quad x \equiv a_2 \pmod{n_2}, \quad \dots \quad x \equiv a_r \pmod{n_r}$$

Find an explicit formula for such a value of x , and find a suitable modulus n in terms of $n_1, n_2, \dots, n_r$ such that any two solutions to the system of congruences are congruent modulo n . $\triangleleft$

 Exercise 7.3.23

Prove that gaps between consecutive primes can be made arbitrarily large. That is, prove that for all $n \in \mathbb{N}$, there exists an integer a such that the numbers

$$a, a+1, a+2, \dots, a+n$$

are all composite. $\triangleleft$

 Exercise 7.3.24

Let $n \in \mathbb{N}$. Prove that n is divisible by 5 if and only if the final digit in the decimal expansion of n is 5 or 0.

More generally, fix $k \geq 1$ and let m be the number whose decimal expansion is given by the last k digits of that of n . Prove that n is divisible by 5^k if and only if m is divisible by 5^k . For example, we have

$$125 \mid 9\,550\,828\,230\,495\,875 \iff 125 \mid 875$$

$\triangleleft$

 Exercise 7.3.25

Let $n \in \mathbb{N}$. Prove that n is divisible by 11 if and only if the *alternating sum* of the digits of n is divisible by 11. That is, prove that if the decimal expansion of n is $d_r d_{r-2} \cdots d_0$, then

$$11 \mid n \iff 11 \mid d_0 - d_1 + d_2 - \cdots + (-1)^r d_r$$

$\triangleleft$

 Exercise 7.3.26

Let $n \in \mathbb{N}$. Find a method for testing if n is divisible by 7 based on the decimal expansion of n . $\triangleleft$

 Exercise 7.3.27

Prove that the RSA algorithm is correct. Specifically, prove:

- (a) If $n = pq$, for distinct positive primes p and q , then $\varphi(n) = (p-1)(q-1)$;
- (b) Given $1 < e < \varphi(n)$ with $e \perp \varphi(n)$, there exists $d \in \mathbb{Z}$ with $de \equiv 1 \pmod{\varphi(n)}$.
- (c) Given $M, K \in \mathbb{Z}$ with $K \equiv M^e \pmod{n}$, it is indeed the case that $K^d \equiv M \pmod{n}$.

$\triangleleft$

Section 7.E

Chapter 7 exercises

Greatest common divisors

In Questions 7.E.1 to 7.E.3, use the Euclidean algorithm (??) to find the greatest common divisor of the given pair of integers.

7.E.1. 382 and 218

7.E.2. 368475 and 26010

7.E.3. 24004512 and 10668672

7.E.4. Let $a, b, c, d \in \mathbb{Z}$ and suppose that $ad - bc = 1$. Prove that $a + b$ and $c + d$ are coprime.

◆ **Definition 7.E.1**

Let $f(x)$ and $g(x)$ be polynomials over $\mathbb{Z}$. We say $f(x)$ **divides** $g(x)$ if there exists a polynomial $q(x)$ over $\mathbb{Z}$ such that $g(x) = q(x)f(x)$.

7.E.5. Prove that $1 - x$ divides $1 - x^2$ and that $1 - x$ does not divide $1 + x^2$.

7.E.6. Prove that $1 + x + x^2$ divides $1 + x^4 + x^5$.

◆ **Definition 7.E.2**

Let $f(x)$ be a polynomial over $\mathbb{Z}$. The **degree** of f , written $\deg(f(x))$, is the greatest power of x in f whose coefficient is nonzero, unless f is the zero polynomial, whose degree is defined to be $-\infty$. We adopt the convention that $(-\infty) + r = -\infty = r + (-\infty)$ for all $r \in \mathbb{N} \cup \{-\infty\}$.

7.E.7. Let $f(x)$ and $g(x)$ be polynomials over $\mathbb{Z}$. Prove that $\deg(f(x), g(x)) = \deg(f(x)) + \deg(g(x))$.

7.E.8. Prove the following modified form of the division theorem (??) for polynomials: given polynomials $f(x)$ and $g(x)$ over $\mathbb{Z}$, with $f(x) \neq 0$, prove that there exist unique polynomials $q(x)$ and $r(x)$ over $\mathbb{Z}$ such that $g(x) = q(x)f(x) + r(x)$ and $\deg(r(x)) < \deg(f(x))$.

Prime numbers

7.E.9. Prove that, for all $k \in \mathbb{N}$, the function $i : \mathbb{N}^k \rightarrow \mathbb{N}$ defined by

$$i(n_1, n_2, \dots, n_k) = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

for all $(n_1, n_2, \dots, n_k) \in \mathbb{N}^k$ is an injection, where $p_1, p_2, p_3, \dots$ is an enumeration of the set of positive primes in increasing order. (Thus $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, and so on.)

7.E.10. Use the result of Question 7.E.9 to construct a bijection

$$\bigcup_{k \in \mathbb{N}} (\mathbb{N}^k \setminus \{(0, 0, \dots, 0)\}) \rightarrow \{n \in \mathbb{N} \mid n \geq 2\}$$

7.E.11. Use a method akin to that of [Question 7.E.9](#) to define an injection $\mathbb{Z}^k \rightarrow \mathbb{N}$.

7.E.12. Define a subset $A \subseteq \mathbb{Z}$ by

$$A = \{n \in \mathbb{N} \mid \exists k > 0, n \mid 12^k - 1\}$$

Find all prime numbers in $\mathbb{Z} \setminus A$.

Base- b expansions

7.E.13. Let $n \in \mathbb{N}$. Prove that the number of trailing 0s in the decimal expansion of $n!$ is equal to

$$\sum_{k=1}^d \left\lfloor \frac{n}{5^k} \right\rfloor$$

where $d \in \mathbb{N}$ is least such that $5^{d+1} > n$, and where $\lfloor x \rfloor$ ([L^AT_EX](#) code: `\lfloor`, `\rfloor`) denotes the floor of x (see ??).

7.E.14. Let $b \in \mathbb{N}$ with $b \geq 2$. Find an expression in terms of $n \in \mathbb{N}$ for the number of trailing 0s in the base- b expansion of $n!$.

True–False questions

In [Questions 7.E.15 to 7.E.20](#), determine (with proof) whether the statement is true or false.

7.E.15. There is an integer that is not coprime to any integer.

7.E.16. Every linear Diophantine equation has a solution.

7.E.17. Every integer n is coprime to its successor $n + 1$.

7.E.18. If the greatest common divisors of two integers is 3, and their least common multiple of 7, then their product is 21.

7.E.19. Every integer is congruent modulo 7 to its remainder when divided by 21.

7.E.20. Let $k \geq 1$. Then $15^k \equiv 1 \pmod{6}$.

Always–Sometimes–Never questions

In [Questions 7.E.21 to 7.E.29](#), determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

7.E.21. Let $a, b, c \in \mathbb{Z}$ and suppose that $\gcd(a, b) = 2$ and $\gcd(b, c) = 2$. Then $\gcd(a, c) = 2$.

7.E.22. Let $a, b \in \mathbb{Z}$. Then $\gcd(a, b) = \gcd(a, \gcd(a, b))$.

7.E.23. Let $a, b \geq 2$. Then $\gcd(a, b) = \text{lcm}(a, b)$.

7.E.24. Let $n \in \mathbb{Z}$. Then $n^2 + 1$ is coprime to $n^4 + n^2 + 1$.

7.E.25. Let $p \in \mathbb{Z}$ be prime. Then p is coprime to all integers $a \neq p$.

7.E.26. Let $p, q, r, s \in \mathbb{Z}$ be positive primes and suppose that $pq = rs$. Then $p = r$ and $q = s$.

7.E.27. Let $p_1, p_2, \dots, p_s, q_1, q_2, \dots, q_t \in \mathbb{Z}$ be prime and suppose that $p_1 p_2 \dots p_s = q_1 q_2 \dots q_t$. Then $s = t$.

7.E.28. Let $a, b \in \mathbb{Z}$ and let p be a positive prime. Then $p \mid (a + b)^p - a - b$.

7.E.29. Let $n \geq 0$ and let $b > 0$. Then n is congruent modulo $b - 1$ to the sum of the digits in its base- b expansion.

Chapter 8

Enumerative combinatorics

Section 8.1

Counting principles

Exercise 8.1.1

Fix $n \in \mathbb{N}$. Prove that $\binom{n}{0} = 1$, $\binom{n}{1} = n$ and $\binom{n}{n} = 1$.



Exercise 8.1.2

List all the permutations of the set $[4]$.



Exercise 8.1.3 (Other characterisations of the factorial)

Let $n \in \mathbb{N}$, and let X and Y be finite sets with $|X| = |Y| = n$.

Prove that the following sets all have size $n!$:

- (a) The set of all n -tuples $(x_0, x_1, \dots, x_{n-1}) \in X^n$ such that each element of X appears exactly once in the list;
- (b) The set of all permutations of X ;
- (c) The set of all enumerations of X (that is, bijections $[n] \rightarrow X$); and
- (d) The set of all bijections $X \rightarrow Y$.



Exercise 8.1.4

Let X and Y be finite sets, and recall that Y^X denotes the set of functions from X to Y . Prove that $|Y^X| = |Y|^{|X|}$.



Exercise 8.1.5

Since September 2001, car number plates on the island of Great Britain have taken the form XX NN XXX, where each X can be any letter of the alphabet except for 'I', 'Q' or 'Z', and NN is the last two digits of the year. [This is a slight simplification of what is really the case, but let's not concern ourselves with *too* many details!] How many possible number plates are there? Number

plates of vehicles registered in the region of Yorkshire begin with the letter ‘Y’. How many Yorkshire number plates can be issued in a given year? $\triangleleft$

 Exercise 8.1.6

Count the number of injections $[3] \rightarrow [4]$. $\triangleleft$

 Exercise 8.1.7

Prove ???. The proof follows the same pattern as that of ??. Be careful to make sure you identify where you use the hypothesis that the sets U_i are pairwise disjoint! $\triangleleft$

 Exercise 8.1.8

Pick your favourite integer $n \geq 1000$. For this value of n , how many inhabited subsets of $[n]$ contain either only even or only odd numbers? (You need not evaluate exponents.) $\triangleleft$

 Exercise 8.1.9

In ??, how many ways could a committee be formed with a *representative* number of people from each colour preference group? That is, the proportion of people on the committee which prefer any of the three colours should be equal to the corresponding proportion of the population of the city. $\triangleleft$

 Exercise 8.1.10

Prove part (b) of ??. $\triangleleft$

 Exercise 8.1.11

Six people are in a room. The atmosphere is tense, since each pair of people is either friends or enemies. There are no allegiances, so for example it is possible for a friend of a friend to be an enemy, or an enemy of a friend to be a friend, and so on. Prove that there is some set of three people that are either all each other’s friends or all each other’s enemies. $\triangleleft$

 Exercise 8.1.12

Make the proof of ?? more formal by defining a bijection between sets of the appropriate sizes. $\triangleleft$

 Exercise 8.1.13

Provide a combinatorial proof that if $n, k \in \mathbb{N}$ with $n \geq k$, then

$$\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$$

Deduce that the combinatorial definition of binomial coefficients (??) is equivalent to the recursive definition (??). $\triangleleft$

 Exercise 8.1.14

Let $n, k \in \mathbb{N}$ with $k \leq n+1$. Prove that

$$k \binom{n}{k} = (n-k+1) \binom{n}{k-1}$$

 Exercise 8.1.15

Given natural numbers n, a, b, c with $a+b+c=n$, define the **trinomial coefficient** $\binom{n}{a,b,c}$ to be the number of ways of partitioning $[n]$ into three sets of sizes a, b and c , respectively. That is,

$\binom{n}{a,b,c}$ is the size of the set

$$\left\{ (A, B, C) \left| \begin{array}{l} A \subseteq [n], \quad B \subseteq [n], \quad C \subseteq [n], \\ |A| = a, \quad |B| = b, \quad |C| = c, \\ \text{and } A \cup B \cup C = [n] \end{array} \right. \right\}$$

By considering trinomial coefficients, prove that if $a, b, c \in \mathbb{N}$, then $(a + b + c)!$ is divisible by $a! \cdot b! \cdot c!$. $\triangleleft$

Section 8.2

Alternating sums

 Exercise 8.2.1

Prove by induction that $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ for all $n \in \mathbb{N}$. ◁

 Exercise 8.2.2

Use ?? to prove that

$$\sum_{k=0}^n (-1)^k \cdot k \cdot \binom{n}{k} = 0$$

for all $n \geq 2$. ◁

 Exercise 8.2.3

Given a set X , prove that the relative complement function $r : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$, defined by $r(U) = X \setminus U$ for all $U \subseteq X$, is an involution. ◁

 Exercise 8.2.4

Prove that every involution is a bijection. ◁

 Exercise 8.2.5

Let $h : X \rightarrow X$ be an involution and let $a \in X$. Prove that h either fixes a —that is, $h(a) = a$ —or swaps it with another element $b \in X$ —that is, $h(a) = b$ and $h(b) = a$. ◁

 Exercise 8.2.6

Let X be a set and let $a \in X$. Prove that the function $T_a : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ defined by $T_a(U) = U \oplus a$ for all $U \subseteq X$ is an involution. ◁

 Exercise 8.2.7

Let X be a finite set and let $a \in X$. Prove that, for all $U \subseteq X$, if $|U|$ is even then $|U \oplus a|$ is odd, and if $|U|$ is odd then $|U \oplus a|$ is even. ◁

 Exercise 8.2.8

Let X be a finite set. Under what conditions does the involution $r : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ given by $r(U) = X \setminus U$ for all $U \subseteq X$ swap parity? ◁

 Exercise 8.2.9

Let $n \in \mathbb{N}$ and let X be the set of all functions $[n] \rightarrow [n]$, partitioned as in ??, so that a function $f : [n] \rightarrow [n]$ has even parity if it fixes an even number of elements, and odd parity if it fixes an odd number of elements. Find a parity-swapping function $X \rightarrow X$. ◁

 Exercise 8.2.10

Repeat [Exercise 8.2.2](#) using the involution principle—that is, use the involution principle to prove that

$$\sum_{k=0}^n (-1)^k \cdot k \cdot \binom{n}{k} = 0$$

for all $n \geq 2$. ◁

 Exercise 8.2.11

Use the involution principle to prove that

$$\sum_{k=0}^n (-1)^k \binom{n}{k} \binom{k}{\ell} = 0$$

for all $n, \ell \in \mathbb{N}$ with $\ell < n$. ◁

 Exercise 8.2.12

Let $n \in \mathbb{N}$ and consider the set

$$X = \{(k, i) \mid k \in [n] \text{ and } i \leq k\}$$

For example, if $n = 3$ then $X = \{(0, 0), (1, 0), (1, 1), (2, 0), (2, 1), (2, 2)\}$.

(a) Prove that $|X| = \sum_{k=0}^n k$.

(b) Use the involution principle to prove that

$$\sum_{k=0}^n (-1)^k k = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$$

◁

 Exercise 8.2.13

Let X, Y, Z be sets. Show that

$$|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$$

Let W be another set. Derive a similar formula for $|W \cup X \cup Y \cup Z|$. ◁

 Exercise 8.2.14

How many natural numbers less than 1000 are multiples of 2, 3, 5 or 7? ◁

Section 8.E

Chapter 8 exercises

Finite sets

8.E.1. Let $n \in \mathbb{N}$ and let $f : [n] \rightarrow [n]$ be a function. Prove that f is injective if and only if f is surjective.

8.E.2. Prove that $|\mathbb{Z}/n\mathbb{Z}| = n$ for all integers $n > 0$.

8.E.3. Let A, B and C be sets. Prove that if $A \triangle B$ and $B \triangle C$ are finite, then $A \triangle C$ is finite, where the notation $\triangle$ refers to the *symmetric difference* operation ([Definition 2.E.1](#))—that is, $A \triangle B = (A \setminus B) \cup (B \setminus A)$, and so on.

Counting

8.E.4. Let X and Y be finite sets with $|X| = m \in \mathbb{N}$ and $|Y| = n \in \mathbb{N}$. Prove that there are 2^{mn} relations from X to Y .

8.E.5. Let X be a set and let R be a relation on X . Prove that R is reflexive if and only if $\Delta_X \subseteq \text{Gr}(R)$, where Δ_X is the diagonal subset of $X \times X$ (see ??). Deduce that if X is finite and $|X| = n \in \mathbb{N}$, then there are $2^{n(n-1)}$ reflexive relations on X .

8.E.6. Let X be a finite set with $|X| = n \in \mathbb{N}$. Prove that there are $2^{\binom{n}{2}} \cdot 2^n$ symmetric relations on X .

8.E.7. Let X be a finite set with $|X| = n \in \mathbb{N}$. Prove that there are $3^{\binom{n}{2}} \cdot 2^n$ antisymmetric relations on X .

8.E.8. Let X be a finite set with $|X| = n \in \mathbb{N}$, let $\sim$ be an equivalence relation on X , and suppose that there is some natural number k such that $|[a]_{\sim}| = k$ for all $a \in X$. Prove that k divides n , and that $|X/\sim| = \frac{n}{k}$.

8.E.9. Let $n, k \in \mathbb{N}$ with $n > 0$ and $k \leq n$. Prove that the number of functions $f : [n] \rightarrow [n]$ that fix exactly k elements of $[n]$ is equal to $\binom{n}{k} (n-1)^{n-k}$.

Double counting

8.E.10. Let $a, b, m, n \in \mathbb{N}$. Prove each of the following by double counting.

(a) $a(m+n) = am + an$

(c) $(a^m)^n = a^{mn}$

(b) $a^{m+n} = a^m \cdot a^n$

(d) $(ab)^n = a^n \cdot b^n$

8.E.11. Prove that $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$ for all $n \in \mathbb{N}$

8.E.12. Prove that $\sum_{k=m}^n \binom{n}{k} \binom{k}{m} = 2^{n-m} \binom{n}{m}$ for all $m, n \in \mathbb{N}$ with $m \leq n$.

8.E.13. Prove that $\sum_{j=0}^k \binom{n-j}{k-j} = \binom{n+1}{k}$ for all $n, k \in \mathbb{N}$ with $k \leq n$.

8.E.14. Prove that $\sum_{k=1}^n \sum_{\ell=0}^k k \binom{n}{k} \binom{n-k}{\ell} = n \cdot 3^{n-1}$ for all $n \in \mathbb{N}$.

8.E.15. Prove that $\binom{n}{r+s+1} = \sum_{k=r+1}^{n-s} \binom{k-1}{r} \binom{n-k}{s}$ for all $n, r, s \in \mathbb{N}$.

8.E.16. Let $a_1, a_2, \dots, a_r \in \mathbb{N}$ and let $n = a_1 + a_2 + \dots + a_r$. Prove that

$$\binom{n}{a_1, a_2, \dots, a_r} = \prod_{k=0}^{r-1} \binom{n - \sum_{i=1}^k a_i}{a_{k+1}}$$

where $\binom{n}{a_1, a_2, \dots, a_r}$ is the number of ordered r -tuples $(U_1, U_2, \dots, U_r)$ such that $U_1, U_2, \dots, U_r$ is a partition of $[n]$ and $|U_k| = a_k$ for all $1 \leq k \leq r$.

Alternating sums

8.E.17. Let X be a finite set. Prove that if $|X|$ is odd then there is no parity-swapping involution $X \rightarrow X$.

8.E.18. Find the number of subsets of $[100]$ that do not contain a multiple of 8.

Miscellaneous exercises

8.E.19. Let $n \in \mathbb{N}$. For each $k \in \mathbb{N}$, find the coefficient of x^k in the polynomial $p(x)$ defined by $p(x) = (1 + x + x^2)^n$.

8.E.20. Fix $n \in \mathbb{N}$. For each $k \in \mathbb{N}$, find the coefficient of x^k in the polynomial $p(x)$ defined by $p(x) = \sum_{i=0}^n x^{n-i} (1+x)^i$.

True–False questions

In Questions 8.E.21 to 8.E.25, determine (with proof) whether the statement is true or false.

8.E.21. Every finite set has a unique enumeration.

8.E.22. The union of a finite number of finite sets is finite.

8.E.23. Every finite set is a proper subset of another finite set.

8.E.24. There is a finite set with infinitely many subsets.

8.E.25. For every partition $\mathcal{A}$ of a finite set X , there is a function $f : X \rightarrow X$ that is a parity-swapping involution with respect to $\mathcal{A}$.

Always–Sometimes–Never questions

In Questions 8.E.26 to 8.E.32, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

8.E.26. Let X and Y be finite sets and let $f : X \rightarrow Y$ be a function. If $|X| \leq |Y|$, then f is injective.

8.E.27. Let X be a finite set. Then there is some proper subset $U \subsetneq X$ such that $|U| = |X|$.

8.E.28. Let X be an infinite set. Then $\mathbb{N} \subseteq X$.

8.E.29. Let X be a finite set. Then $\mathcal{P}(X)$ is finite.

8.E.30. Let $a, b \in \mathbb{R}$ with $a < b$. Then the set $[a, b] \cap \mathbb{Q}$ is finite.

8.E.31. Let X and Y be sets with X finite, and let $f : X \rightarrow Y$ be an injection. Then Y is finite.

8.E.32. Let X and Y be sets with X finite, and let $f : X \rightarrow Y$ be a surjection. Then Y is finite.

Chapter 9

Real numbers

Section 9.1

Inequalities and means

 Exercise 9.1.1

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Prove that $\|\vec{x} - \vec{y}\| = \|\vec{y} - \vec{x}\|$. That is, the distance from $\vec{x}$ to $\vec{y}$ is equal to the distance from $\vec{y}$ to $\vec{x}$. ◁

 Exercise 9.1.2

Prove that if $x \in \mathbb{R}$ then the magnitude $\|(x)\|$ is equal to the absolute value $|x|$. ◁

 Exercise 9.1.3

Let $\vec{x} \in \mathbb{R}^n$. Prove that $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$. ◁

 Exercise 9.1.4

Let $n \in \mathbb{N}$ and let $x_i \in \mathbb{R}$ for each $i \in [n]$. Prove that

$$\left| \sum_{i=1}^n x_i \right| \leq \sum_{i=1}^n |x_i|$$

with equality if and only if the numbers x_i are either all non-positive or all non-negative. ◁

 Exercise 9.1.5

Let $x, y \in \mathbb{R}$. Prove that

$$||x| - |y|| \leq |x - y|$$

 Exercise 9.1.6

Let $\vec{x}, \vec{y}, \vec{z}, \vec{w} \in \mathbb{R}^n$ and let $a, b, c, d \in \mathbb{R}$. Prove that

$$(a\vec{x} + b\vec{y}) \cdot (c\vec{z} + d\vec{w}) = ac(\vec{x} \cdot \vec{z}) + ad(\vec{x} \cdot \vec{w}) + bc(\vec{y} \cdot \vec{z}) + bd(\vec{y} \cdot \vec{w})$$

 Exercise 9.1.7

Let $r \in \mathbb{N}$ and let $a_1, a_2, \dots, a_r \in \mathbb{R}$ be such that $a_1^2 + a_2^2 + \dots + a_n^2 = 6$. Prove that

$$(a_1 + 2a_2 + \dots + na_n)^2 \leq n(n+1)(2n+1)$$

and determine when equality holds. ◁

 Exercise 9.1.8

Let $a, b > 0$ be real numbers. Prove that $\frac{a^2 + b^2}{2} \geq ab$. ◁

 Exercise 9.1.9

Let $x > 0$ and let $n \in \mathbb{N}$. Find the minimum possible value of $\sum_{k=-n}^n x^k$. ◁

 Exercise 9.1.10

Let $r \in \mathbb{N}$ and let $x_1, x_2, \dots, x_{2r} \in \mathbb{R}$. Write

$$a = \frac{1}{r} \sum_{i=1}^r x_i \quad \text{and} \quad g = \sqrt[r]{\prod_{i=1}^r x_i}$$

for the arithmetic and geometric means, respectively, of the numbers $x_1, \dots, x_r$; write

$$a' = \frac{1}{r} \sum_{i=r+1}^{2r} x_i \quad \text{and} \quad g' = \sqrt[r]{\prod_{i=r+1}^{2r} x_i}$$

for the arithmetic and geometric means, respectively, of the numbers $x_{r+1}, \dots, x_{2r}$; and write

$$A = \frac{1}{2r} \sum_{i=1}^{2r} x_i \quad \text{and} \quad G = \sqrt[2r]{\prod_{i=1}^{2r} x_i}$$

for the arithmetic and geometric means, respectively, of all the numbers $x_1, \dots, x_{2r}$.

Prove that

$$A = \frac{a + a'}{2} \quad \text{and} \quad G = \sqrt{gg'}$$

Deduce that, for each $r \in \mathbb{N}$, if $p_{\text{AM-GM}}(r)$ is true then $p_{\text{AM-GM}}(2r)$ is true. Deduce further than $p_{\text{AM-GM}}(2^m)$ is true for all $m \in \mathbb{N}$. ◁

 Exercise 9.1.11

Let $r \geq 2$ and let $x_1, \dots, x_{r-1} \in \mathbb{N}$. Define

$$x_r = \frac{1}{r-1} \sum_{i=1}^{r-1} x_i$$

Prove that

$$\frac{1}{r} \sum_{i=1}^r x_i = x_r$$

Assuming $p_{\text{AM-GM}}(r)$, deduce that

$$x_r^r \geq \prod_{i=1}^r x_i = \left(\prod_{i=1}^{r-1} x_i \right) \cdot x_r$$

with equality if and only if $x_1 = x_2 = \dots = x_r$. Deduce that $p_{\text{AM-GM}}(r)$ implies $p_{\text{AM-GM}}(r-1)$. Use [Exercise 9.1.10](#) to deduce further that $p_{\text{AM-GM}}(n)$ is true for all $n \geq 1$. ◁

 Exercise 9.1.12

Prove the GM–HM inequality for general values of $n \in \mathbb{N}$.

◁

 Exercise 9.1.13

Let $n > 0$ and let $x_1, x_2, \dots, x_n$ be positive real numbers. Prove that

$$\min\{x_1, x_2, \dots, x_n\} \leq \left(\frac{1}{n} \sum_{i=1}^n x_i^{-1} \right)^{-1} \quad \text{and} \quad \max\{x_1, x_2, \dots, x_n\} \geq \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{\frac{1}{2}}$$

with equality in each case if and only if $x_1 = x_2 = \dots = x_n$.

◁

Section 9.2

Completeness and convergence

 Exercise 9.2.1

Define the notions of **lower bound** and **infimum**, and find the infimum of the open interval $(0, 1)$.

◁

 Exercise 9.2.2

Let (x_n) be the constant sequence with value $a \in \mathbb{R}$. Prove that $(x_n) \rightarrow a$.

◁

 Exercise 9.2.3

Prove that the sequence (z_n) defined by $z_n = \frac{n+1}{n+2}$ converges to 1.

◁

 Exercise 9.2.4

Let (y_n) be the sequence defined by $y_n = n$ for all $n \in \mathbb{N}$:

$$(0, 1, 2, 3, \dots)$$

Prove that (y_n) diverges.

◁

 Exercise 9.2.5

Let $r \in \mathbb{R}$. Recall that $(r^n) \rightarrow 0$ if $|r| < 1$ (this was ??) and that (r^n) diverges if $r = -1$ (this was ??). Prove that (r^n) diverges if $|r| > 1$.

◁

 Exercise 9.2.6

Prove that if a sequence (x_n) converges to a nonzero limit, then (x_n) is eventually nonzero. Find a sequence (x_n) that converges to zero, but is not eventually nonzero.

◁

 Exercise 9.2.7

Let (x_n) be a sequence and let $p(x)$ be a logical formula. What does it mean to say that $p(x_n)$ is *not* eventually true? Find a sentence involving the phrase ‘not eventually’ that is equivalent to the assertion that (x_n) diverges.

◁

 Exercise 9.2.8 (‘Eventually’ does not preserve negation)

Find a sequence (x_n) and a logical formula $p(x)$ such that $p(x_n)$ is neither eventually true nor eventually false. (Thus ‘ $p(x_n)$ is not eventually true’ does not imply ‘ $\neg p(x_n)$ is eventually true’.)

◁

 Exercise 9.2.9

Prove parts (b) and (d) of ??.

◁

 Exercise 9.2.10

Let (x_n) be a sequence of real numbers converging to a real number a , and let $p(x) = a_0 + a_1x + \dots + a_d x^d$ be a polynomial function. Prove that $(p(x_n)) \rightarrow p(a)$, and that $\left(\frac{1}{p(x_n)}\right) \rightarrow \frac{1}{p(a)}$ if $p(a) \neq 0$.

◁

 Exercise 9.2.11

Fix $r \in \mathbb{N}$, and let $p(x) = a_0 + a_1x + \dots + a_r x^r$ and $q(x) = b_0 + b_1x + \dots + b_r x^r$ be polynomials with real coefficients. Prove that if $b_r \neq 0$, then $\left(\frac{p(n)}{q(n)}\right) \rightarrow \frac{a_r}{b_r}$.

◁

 Exercise 9.2.12

Prove that the sequence $(5^n - n^5)_{n \geq 0}$ is *eventually* increasing—that is, there is some $k \in \mathbb{N}$ such that $(5^n - n^5)_{n \geq k}$ is an increasing sequence.

◁

 Exercise 9.2.13

Prove part (b) of the monotone convergence theorem (??). That is, prove that if a sequence (x_n) is decreasing and has a lower bound, then it converges. $\triangleleft$

 Exercise 9.2.14

Use the monotone convergence theorem to prove that the sequence $\left(\frac{n!}{n^n}\right)$ converges. $\triangleleft$

 Exercise 9.2.15

A sequence (x_n) is defined recursively by $x_0 = 0$ and $x_{n+1} = \sqrt{2 + x_n}$ for all $n \geq 0$. That is,

$$x_n = \underbrace{\sqrt{2 + \sqrt{2 + \sqrt{\cdots + \sqrt{2}}}}}_{n \text{ '2's'}}$$

Prove that (x_n) converges. $\triangleleft$

 Exercise 9.2.16

Prove that a subsequence of an increasing sequence is increasing, that a subsequence of a decreasing sequence is decreasing, and that a subsequence of a constant sequence is constant. $\triangleleft$

 Exercise 9.2.17

Prove that every convergent sequence is a Cauchy sequence. $\triangleleft$

Section 9.3

Series and sums

 Exercise 9.3.1

Prove that the series $\sum_{n \geq 3} \binom{n}{3}^{-1}$ converges, and find its sum. ◁

 Exercise 9.3.2

Prove that the series $\sum_{n \geq 0} (-1)^n$ diverges. ◁

 Exercise 9.3.3

Prove that the series $\sum_{n \geq 0} r^n$ diverges for all $r \in \mathbb{R} \setminus (-1, 1)$. ◁

 Exercise 9.3.4

Prove part (b) of ??, and deduce that if $\sum_{n \geq 0} a_n$ and $\sum_{n \geq 0} b_n$ are convergent series, then $\sum_{n \geq 0} (a_n - b_n)$ converges, and its sum is equal to $\sum_{n \geq 0} a_n - \sum_{n \geq 0} b_n$. ◁

 Exercise 9.3.5

Prove the ‘uniqueness’ part of ??—that is, prove that for all $x \in [0, \infty)$, if

$$\sum_{i \geq 0} \frac{u_i}{b^i} \quad \text{and} \quad \sum_{i \geq 0} \frac{v_i}{b^i}$$

are two series satisfying conditions (i)–(iii) of ??, then $u_i = v_i$ for all $i \in \mathbb{N}$. ◁

 Exercise 9.3.6

Use ?? to find the decimal expansion of $\frac{1}{6}$. ◁

 Exercise 9.3.7

Use ?? to find the *binary* expansion of $\frac{1}{7}$. ◁

 Exercise 9.3.8

Prove that between any two distinct real numbers, there is a rational number. ◁

 Exercise 9.3.9

Let $a \in \mathbb{R}$. Prove that the series $\sum_{n \geq 0} a$ converges if and only if $a = 0$. ◁

 Exercise 9.3.10

Prove part (b) of ??. ◁

 Exercise 9.3.11

Prove that $\sum_{n \geq 1} n^{-r}$ diverges for all real $r \geq 1$. ◁

 Exercise 9.3.12

Prove that if (a_n) is a sequence such that $(a_n) \rightarrow 0$ and $a_n \leq 0$ for all $n \in \mathbb{N}$. Prove that if (a_n) is an increasing sequence, then the series $\sum_{n \geq 0} (-1)^n a_n$ converges. ◁

 Exercise 9.3.13

Find a decreasing sequence (a_n) of non-negative real numbers such that $\sum_{n \geq 0} (-1)^n a_n$ diverges. ◁

 Exercise 9.3.14Find a series that converges, but does not converge absolutely. ◁ **Exercise 9.3.15**Prove ?? with ‘convergent’ replaced by ‘absolutely convergent’ throughout. ◁ **Exercise 9.3.16**Prove part (b) of ??. ◁ **Exercise 9.3.17**Use the ratio test to prove that the series $\sum_{n \geq 0} \frac{x^n}{n!}$ converges for all $x \in \mathbb{R}$. ◁ **Exercise 9.3.18**Let $J = \{j_n \mid n \in \mathbb{N}\}$ be a set such that, for all $m, n \in \mathbb{N}$, we have $j_m \neq j_n$. Let $(a_j)_{j \in J}$ be a J -indexed sequence such that either (i) $\sum_{j \in J} a_j$ is absolutely convergent, or (ii) $a_j \geq 0$ for all $j \in J$. Prove that ifthere is a bijection $\sigma : I \rightarrow J$, then $\sum_{i \in I} a_{\sigma(i)} = \sum_{j \in J} a_j$. ◁ **Exercise 9.3.19**Let $x \in (-1, 1)$. Prove that $\sum_{n \geq 1} nx^{n-1} = \frac{1}{(1-x)^2}$. ◁ **Exercise 9.3.20**Prove that $\exp(x) = e^x$ for all $x \in \mathbb{Q}$. ◁

Section 9.E

Chapter 9 exercises

Inequalities and means

9.E.1. Compute $\|2(-1, -5, 1, 5, 1) - (2, -1, 1, -2, 8)\|$.

9.E.2. Let $a, b \in \mathbb{R}$ and let $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$ with $a_0 = a$ and $a_n = b$. Prove that $|a - b| \leq \sum_{k=1}^n |a_{k-1} - a_k|$.

9.E.3. Let $n \in \mathbb{N}$. Prove that $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$ for all $\vec{x}, \vec{y} \in \mathbb{R}^n$.

9.E.4. Let $n \in \mathbb{N}$, let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and let $a, b, c, d \in \mathbb{R}$. Prove that $(a\vec{x} + b\vec{y}) \cdot (c\vec{x} + d\vec{y}) = ac\|\vec{x}\|^2 + bd\|\vec{y}\|^2 + (ad + bc)(\vec{x} \cdot \vec{y})$.

9.E.5. Let $a, b, c, d \in \mathbb{R}$. Assume that $a^2 + b^2 = 9$, $c^2 + d^2 = 25$ and $ac + bd = 15$. Find the value of $\frac{a+2b}{c+2d}$.

9.E.6. Let $\vec{a} \in \mathbb{R}^3$ and $r > 0$, and define

$$B(\vec{a}; r) = \{\vec{x} \in \mathbb{R}^3 \mid \|\vec{x} - \vec{a}\| < r\}$$

Prove that $\|\vec{x} - \vec{y}\| < 2r$ for all $\vec{x}, \vec{y} \in B(\vec{a}; r)$.

9.E.7. Prove that $a^3 + b^3 + c^3 \geq abc$ for all positive real numbers a, b, c . When does equality occur?

9.E.8. Let $a, b, c > 0$. Prove that $\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \geq 6$, with equality if and only if $a = b = c$.

9.E.9. Let $a, b, c \in \mathbb{R}$. Prove that $a^4 + b^4 + c^4 \geq \frac{(ab + bc + ca)^2}{3}$. When does equality occur?

9.E.10. *Pound cost averaging* is an investment strategy in which an investor purchases asset in equal instalments over a fixed period of time.

Let $a, b \in \mathbb{R}$ with $a < b$, let $n \geq 1$, and let $f : [a, b] \rightarrow (0, \infty)$ be such that at time $t \in [a, b]$, a stock is trading at $\check{c}f(t)$ per share. Assume that you purchase a total of $\check{c}M$ of the stock in equal instalments of $\check{c}\frac{M}{n}$ at times $t_0, t_1, \dots, t_{n-1}$, where $M \in [0, \infty)$, $n \geq 1$, and $t_i = a + i \cdot \frac{b-a}{n}$ for all $0 \leq i < n$.

(a) Prove that the value of the shares that you hold at time b is equal to

$$\check{c} \frac{f(b)}{\text{HM}(f(t_0), f(t_1), \dots, f(t_{n-1}))} \cdot M$$

where $\text{HM}(a_1, a_2, \dots, a_n)$ denotes the harmonic mean of $a_1, a_2, \dots, a_n \in \mathbb{R}$.

(b) Under what condition have you made a profit?

Sequences

9.E.11. Does there exist a sequence (x_n) such that $(x_{n+1} - x_n) \rightarrow 0$ but (x_n) diverges?

True–False questions

In Questions 9.E.12 to 9.E.18, determine (with proof) whether the statement is true or false.

9.E.12. For all $x, y \in \mathbb{R}$ we have $|x - y| \leq |x| - |y|$.

9.E.13. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $f(\vec{x}) = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$ is injective.

9.E.14. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $f(\vec{x}) = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$ is surjective.

9.E.15. For all $\vec{x}, \vec{y} \in \mathbb{R}^n$ we have $\|2\vec{x} - 3\vec{y}\| \leq 2\|\vec{x}\| + 3\|\vec{y}\|$.

9.E.16. Every convergent sequence is bounded.

9.E.17. Every convergent sequence is eventually monotone.

9.E.18. Every subsequence of a divergent sequence diverges.

Always–Sometimes–Never questions

In Questions 9.E.19 to 9.E.26, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

9.E.19. Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and suppose that $\|\vec{x} - \vec{y}\| = 0$. Then $\vec{x} = \vec{y}$.

9.E.20. Let $x_1, x_2, \dots, x_n \geq 0$. Then the harmonic mean of the x_i s is greater than their quadratic mean.

9.E.21. Let $y_1, y_2, \dots, y_m \geq 0$. Then the geometric mean of the y_j s is less than or equal to their quadratic mean.

9.E.22. Let (x_n) be a sequence of real numbers and suppose that the subsequences (x_{2n}) and (x_{2n+1}) both converge. Then (x_n) converges.

9.E.23. Let (x_n) be a sequence of real numbers and suppose that the subsequences (x_{2n}) , (x_{2n+1}) and (x_{3n}) all converge. Then (x_n) converges.

9.E.24. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, let (x_n) be a sequence of real numbers, and suppose that $(x_n) \rightarrow a \in \mathbb{R}$. Then $(x_n^2) \rightarrow a^2$.

9.E.25. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, let (x_n) be a sequence of real numbers, and suppose that $(x_n) \rightarrow a \in \mathbb{R}$. Then $(f(x_n)) \rightarrow f(a)$.

9.E.26. Let (a_n) be a sequence of real numbers and suppose that $(a_n) \rightarrow 0$. Then $\sum_{n \geq 0} a_n$ converges.

Chapter 10

Infinity

Section 10.1

Cardinality

 Exercise 10.1.1

Consider the function $f : \mathbb{R} \rightarrow (-1, 1)$ defined by

$$f(x) = \frac{x}{1 + |x|}$$

for all $x \in \mathbb{R}$. Prove that f is a bijection, and deduce that $|(-1, 1)| = \mathfrak{c}$. ◁

 Exercise 10.1.2

Let $a, b \in \mathbb{R}$ with $a < b$. Prove that each of the sets (a, b) , $[a, b)$, $(a, b]$ and $[a, b]$ has cardinality $\mathfrak{c}$. ◁

 Exercise 10.1.3

Prove that $n < \aleph_0$ for all $n \in \mathbb{N}$. ◁

 Exercise 10.1.4

Prove that $\leq$ is a reflexive, transitive relation on Card . That is, prove that:

(a) $\kappa \leq \kappa$ for all cardinal numbers κ ; and

(b) For all cardinal numbers κ, λ, μ , if $\kappa \leq \lambda$ and $\lambda \leq \mu$, then $\kappa \leq \mu$. ◁

 Exercise 10.1.5

Prove that $\sim$ is an equivalence relation on $[\kappa]$. ◁

 Exercise 10.1.6

Prove that p and q are well-defined, and that q is an inverse for p . ◁

 Exercise 10.1.7

Prove that each of the functions defined in the above three cases is a bijection. ◁

 Exercise 10.1.8

Let $\mathcal{F}(\mathbb{N})$ be the set of all finite subsets of $\mathbb{N}$, and define $f : \mathcal{F}(\mathbb{N}) \rightarrow \mathbb{N}$ by

$$f(U) = \sum_{a \in U} 10^a = \sum_{k=1}^n 10^{a_k}$$

for all $U = \{a_1, a_2, \dots, a_n\} \in \mathcal{F}(\mathbb{N})$. Put another way, $f(U)$ is the natural number whose decimal expansion has a 1 in the r^{th} position (counting from $r = 0$) if $r \in U$, and a 0 otherwise. For example

$$f(\{0, 1, 3, 4, 8\}) = 100011011 \quad \text{and} \quad f(\emptyset) = 0$$

Prove that f is injective, and use the Cantor–Schröder–Bernstein theorem to deduce that $\mathcal{F}(\mathbb{N})$ is countably infinite. $\triangleleft$

 Exercise 10.1.9

Let X , Y and Z be sets. Prove that if $|X| = |Z|$ and $X \subseteq Y \subseteq Z$, then $|X| = |Y| = |Z|$. $\triangleleft$

Section 10.2

Cardinal arithmetic

 Exercise 10.2.1

Prove parts (b) and (c) of ??.

◁

 Exercise 10.2.2Prove that $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$.

◁

 Exercise 10.2.3Let X be a set and let $\sim$ be an equivalence relation on X . Prove that if κ is a cardinal number such that $|[a]_{\sim}| = \kappa$ for all $a \in X$, then $|X| = |X/\sim| \cdot \kappa$.

◁

 Exercise 10.2.4

Prove parts (b), (c) and (d) of ??.

◁

 Exercise 10.2.5Prove that for all cardinal numbers κ, λ, μ , if $\lambda \leq \mu$, then $\lambda^\kappa \leq \mu^\kappa$.

◁

 Exercise 10.2.6

Prove parts (b) and (c) of ??.

◁

 Exercise 10.2.7Prove that $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$.

◁

 Exercise 10.2.8Prove that $\aleph_0^{\aleph_0^{\aleph_0}} = 2^{\mathfrak{c}}$.

◁

 Exercise 10.2.9Let $\{X_i \mid i \in I\}$ be a family of sets indexed by a set I , and define a function

$$q : \bigsqcup_{i \in I} X_i \rightarrow \bigcup_{i \in I} X_i$$

by $q(i, a) = a$ for all $i \in I$ and $a \in X_i$. Prove that if the sets X_i for $i \in I$ are pairwise disjoint, then q is a bijection.

◁

 Exercise 10.2.10Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of natural numbers, and let $I = \{n \in \mathbb{N} \mid a_n > 0\}$. Prove that

$$\sum_{n \in \mathbb{N}} a_n = \begin{cases} \aleph_0 & \text{if } I \text{ is infinite} \\ \sum_{k=1}^n a_{n_k} & \text{if } I = \{n_k \mid k \in [n]\} \text{ is finite} \end{cases}$$

 Exercise 10.2.11Prove that $1 \times 2 \times 3 \times 4 \times \cdots = 2^{\aleph_0}$.

◁

 Exercise 10.2.12Prove that there do not exist cardinal numbers $\{\kappa_n \mid n \in \mathbb{N}\}$ such that $\kappa_n \neq 1$ for all $n \in \mathbb{N}$ and $\prod_{n \in \mathbb{N}} \kappa_n = \aleph_0$.

◁

Section 10.E

Chapter 10 exercises

Countability

In Questions 10.E.1 to 10.E.4, prove that the set is countable.

10.E.1. The set $D = \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\}$ of *dyadic* rational numbers.

10.E.2. The set of all functions $[n] \rightarrow \mathbb{Z}$, where $n \in \mathbb{N}$.

10.E.3. The set of all real numbers whose square is rational.

10.E.4. The following set:

$$[(\mathbb{Z} \times \mathbb{Q}) \setminus (\mathbb{N} \times \mathbb{Z})] \cup \{n \in \mathbb{N} \mid \exists u, v \in \mathbb{N}, n = 5u + 6v\} \cup \{x \in \mathbb{R} \mid x - \sqrt{2} \in \mathbb{Q}\}$$

In Questions 10.E.5 to 10.E.7, prove that the set is uncountable.

10.E.5. The set of all functions $\mathbb{Z} \rightarrow [2]$.

10.E.6. The set of all subsets $U \subseteq \mathbb{N}$ such that neither U nor $\mathbb{N} \setminus U$ is finite.

10.E.7. The set of all sequences of rational numbers that converge to 0.

In Questions 10.E.8 to 10.E.12, determine whether the set is countable or uncountable, and then prove it.

10.E.8. The set of all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ that are weakly decreasing—that is, such that for all $m, n \in \mathbb{N}$, if $m \leq n$, then $f(m) \geq f(n)$.

10.E.9. The set of all functions $f : \mathbb{N} \rightarrow \mathbb{Z}$ that are weakly decreasing.

10.E.10. The set of all periodic functions $f : \mathbb{Z} \rightarrow \mathbb{Q}$ —that is, such that there is some integer $p > 0$ such that $f(x+p) = f(x)$ for all $x \in \mathbb{Z}$.

10.E.11. The set of all periodic functions $f : \mathbb{Q} \rightarrow \mathbb{Z}$ —that is, such that there is some rational number $p > 0$ such that $f(x+p) = f(x)$ for all $x \in \mathbb{Q}$.

10.E.12. The set of all real numbers x such that $a_0 + a_1x + \cdots + a_dx^d = 0$ for some rational numbers $a_0, a_1, \dots, a_d$ with $a_d \neq 0$.

10.E.13. A subset $D \subseteq \mathbb{R}$ is *dense* if $(a - \varepsilon, a + \varepsilon) \cap D$ is inhabited for all $a \in \mathbb{R}$ and all $\varepsilon > 0$ —intuitively, this means that there are elements of D arbitrarily close to any real number. Must a dense subset of $\mathbb{R}$ be uncountable?

Cardinality

In Questions 10.E.14 to 10.E.20, determine whether the cardinality of the set is equal to $\aleph_0$, equal to $\mathfrak{c}$, or greater than $\mathfrak{c}$.

10.E.14. $\mathcal{P}(\mathbb{R})$.

10.E.15. The set of all finite subsets of $\mathbb{R}$.

10.E.16. The set of all countably infinite subsets of $\mathbb{R}$.

10.E.17. $\mathbb{R} \times \mathbb{R}$.

10.E.18. The set of all functions $f : \mathbb{N} \rightarrow \mathbb{R}$ such that $f(n) = 0$ for all but finitely many $n \in \mathbb{N}$.

10.E.19. The set of all functions $\mathbb{Q} \rightarrow \mathbb{R}$.

10.E.20. The set of all functions $\mathbb{R} \rightarrow \mathbb{R}$.

10.E.21. Prove that there is a set $\mathcal{C}$ of pairwise disjoint circles in $\mathbb{R}^2$ such that $|\mathcal{C}| = \mathfrak{c}$; formally, a circle is a subset $C \subseteq \mathbb{R}^2$ of the form

$$C = \{(x, y) \mid (x - a)^2 + (y - b)^2 = r^2\}$$

for some $a, b \in \mathbb{R}$ and some $r > 0$.

10.E.22. Prove that there does not exist a set $\mathcal{D}$ of pairwise disjoint discs in $\mathbb{R}^2$ such that $|\mathcal{D}| = \mathfrak{c}$; formally, a disc is a subset $D \subseteq \mathbb{R}^2$ of the form

$$D = \{(x, y) \mid (x - a)^2 + (y - b)^2 \leq r^2\}$$

for some $a, b \in \mathbb{R}$ and some $r > 0$.

Cardinal arithmetic

10.E.23. Prove that $\kappa + 0 = \kappa \cdot 1 = \kappa^1 = \kappa$ for all cardinal numbers κ .

10.E.24. Prove that $\kappa^0 = 1^\kappa = 1$ for all cardinal numbers κ .

10.E.25. Let κ be an infinite cardinal number such that $\kappa \cdot \kappa = \kappa$. Prove that $\kappa^\kappa = 2^\kappa$.

10.E.26. Prove that $\mathfrak{c}^\mathfrak{c} = 2^{2^{\aleph_0}}$.

10.E.27. Let κ , λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leq \lambda$, then $\kappa + \mu \leq \lambda + \mu$.

(b) Suppose that $\kappa + \mu \leq \lambda + \mu$. Must it be the case that $\kappa \leq \lambda$?

10.E.28. Let κ , λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leq \lambda$, then $\kappa \cdot \mu \leq \lambda \cdot \mu$.

(b) Suppose that $\kappa \cdot \mu \leq \lambda \cdot \mu$ and $\mu > 0$. Must it be the case that $\kappa \leq \lambda$?

10.E.29. Let κ , λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leq \lambda$, then $\kappa^\mu \leq \lambda^\mu$.

(b) Suppose that $\kappa^\mu \leq \lambda^\mu$ and $\mu > 0$. Must it be the case that $\kappa \leq \lambda$?

10.E.30. Let κ , λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leq \lambda$, then $\mu^\kappa \leq \mu^\lambda$.

(b) Suppose that $\mu^\kappa \leq \mu^\lambda$ and $\mu > 1$. Must it be the case that $\kappa \leq \lambda$?

◆ **Definition 10.E.1**

Given cardinal numbers κ and λ , define the **binomial coefficient** $\binom{\kappa}{\lambda}$ by

$$\binom{\kappa}{\lambda} = |\{U \subseteq [\kappa] \mid |U| = \lambda\}|$$

10.E.31. Let κ be a cardinal number. Prove that

$$\binom{\aleph_0}{\kappa} = \begin{cases} 1 & \text{if } \kappa = 0 \\ \aleph_0 & \text{if } \kappa \in \mathbb{N} \text{ and } \kappa > 0 \\ 2^{\aleph_0} & \text{if } \kappa = \aleph_0 \\ 0 & \text{if } \kappa \notin \mathbb{N} \cup \{\aleph_0\} \end{cases}$$

10.E.32. Find the values of $\binom{c}{\kappa}$ for $\kappa \in \mathbb{N} \cup \{\aleph_0, c\}$.

10.E.33. Prove that $\sum_{\lambda \leq \kappa} \binom{\kappa}{\lambda} = 2^\kappa$ for all cardinal numbers κ .

10.E.34. Define the factorial $\kappa!$ of a cardinal number κ in a way that generalises the notion of factorial $n!$ for a natural number n . Find expressions for $\aleph_0!$ and $c!$.

True–False questions

In Questions 10.E.35 to 10.E.45, determine (with proof) whether the statement is true or false.

10.E.35. For any two countably infinite sets X and Y , there exists a bijection $X \rightarrow Y$.

10.E.36. For any two uncountable sets X and Y , there exists a bijection $X \rightarrow Y$.

10.E.37. The product of countably many countable sets is countable.

10.E.38. Every countably infinite set can be partitioned into infinitely many infinite subsets.

10.E.39. There are countably many infinite subsets of $\mathbb{N}$.

10.E.40. The set $\mathbb{N} \times \mathbb{Z} \times \mathbb{Q}$ is countably infinite.

10.E.41. $\aleph_0$ is the smallest cardinal number.

10.E.42. There exist sets X and Y of different cardinalities that have the same number of finite subsets.

10.E.43. There exist sets X and Y of different cardinalities that have the same number of countably infinite subsets.

10.E.44. For all sets X , if $U \subsetneq X$, then $|U| < |X|$.

10.E.45. $2^\kappa < 3^\kappa$ for all cardinal numbers $\kappa > 0$.

Always–Sometimes–Never questions

In Questions 10.E.46 to 10.E.51, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

10.E.46. Let X be a set and suppose that there exists an injection $\mathbb{N} \rightarrow X$. Then there exists an injection $\mathbb{Q} \rightarrow X$.

10.E.47. Let X be a set. Then $\mathcal{P}(X)$ is countably infinite.

10.E.48. Let X be a set, let κ and λ be cardinal numbers, and suppose that $\lambda \leq \kappa = |X|$. Then X has a subset of size λ .

10.E.49. Let X and Y be sets with $|X| \leq |Y|$. Then $|\mathcal{P}(X)| \leq |\mathcal{P}(Y)|$.

10.E.50. Let $n \in \mathbb{N}$ with $n \geq 2$. Then $\aleph_0^n > \aleph_0$.

10.E.51. Let κ be a cardinal number with $\kappa \geq \aleph_0$. Then $\aleph_0^\kappa > \aleph_0$.

Trick question

10.E.52. Does there exist a cardinal number κ such that $\aleph_0 < \kappa < 2^{\aleph_0}$?

Chapter 11

Discrete probability theory

Section 11.1

Discrete probability spaces

 Exercise 11.1.1

A fair six-sided die is rolled three times. What is the probability that the sum of the die rolls is less than or equal to 12, given that each die roll shows a power of 2? ◁

 Exercise 11.1.2

Let $(\Omega, \mathbb{P})$ be a probability space and let A, B be events with $\mathbb{P}(B) > 0$. Prove that

$$\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B \mid B)$$
◁

 Exercise 11.1.3

Let $(\Omega, \mathbb{P})$ be a probability space and let A, B be events such that $\mathbb{P}(B) > 0$. Prove that $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ if and only if A and B are independent. ◁

 Exercise 11.1.4

In ??, find the probabilities that the car was an Allegheny and that the car was a Monongahela. ◁

Section 11.2

Discrete random variables

 Exercise 11.2.1

With probability space $(\Omega, \mathbb{P})$ and random variable N defined as in ??, compute $\mathbb{P}\{N \text{ is odd}\}$ and $\mathbb{P}\{N = 4\}$. ◁

 Exercise 11.2.2

Let $(\Omega, \mathbb{P})$ be a probability space, let E be a set, let X be an E -valued random variable and let $U \subseteq E$. Prove that the event $\{X \in U\}$ is equal to the preimage $X^{-1}[U]$. Deduce that

$$\mathbb{P}\{X \in U\} = \sum_{e \in U} f_X(e)$$
◁

 Exercise 11.2.3

A coin which shows heads with probability $p \in [0, 1]$, and tails otherwise, is flipped five times. For each $1 \leq i \leq 5$, let

$$X_i = \begin{cases} 0 & \text{if the } i^{\text{th}} \text{ flip shows tails} \\ 1 & \text{if the } i^{\text{th}} \text{ flip shows heads} \end{cases}$$

Prove that the random variables X_1, X_2, X_3, X_4, X_5 are pairwise independent. ◁

 Exercise 11.2.4

Let $(\Omega, \mathbb{P})$ be the probability space modelling the roll of a fair six-sided die, and let X be the $\{0, 1\}$ -valued random variable which is equal to 0 if the die shows an even number and 1 if the die shows an odd number. Prove that $X \sim \text{Unif}(\{0, 1\})$. ◁

 Exercise 11.2.5

Let X be a $\{0, 1\}$ -valued random variable. Prove that $X \sim \text{U}(\{0, 1\})$ if and only if $X \sim \text{B}(1, \frac{1}{2})$. ◁

 Exercise 11.2.6

Let $p \in [0, 1]$ and let $X \sim \text{Geom}(p)$. Prove that

$$\mathbb{P}\{X \text{ is even}\} = \frac{1}{2-p}$$

What is the probability that X is odd? ◁

 Exercise 11.2.7

Let $(\Omega, \mathbb{P})$ be a probability space and let $c \in \mathbb{R}$. Thinking of c as a *constant* real-valued random variable,^[a] prove that $\mathbb{E}[c] = c$. ◁

 Exercise 11.2.8

Use ?? to prove that the expectation of a $\mathbb{N}$ -valued random variable which is geometrically distributed with parameter $p \in [0, 1]$ is equal to $\frac{1-p}{p}$. Use this to compute the expected number of times a coin must be flipped before the first time heads shows, given that heads shows with probability $\frac{2}{7}$. ◁

 Exercise 11.2.9

Prove that the expectation of a $\mathbb{N}^+$ -valued random variable which is geometrically distributed with parameter $p \in [0, 1]$ is equal to $\frac{1}{p}$. ◁

^[a]Formally, we should define $X : \Omega \rightarrow \mathbb{R}$ by letting $X(\omega) = c$ for all $\omega \in \Omega$; then compute $\mathbb{E}[X]$.

 Exercise 11.2.10

Let $(\Omega, \mathbb{P})$ be a probability space, let $E \subseteq \mathbb{R}$ be countable, let $\{X_i \mid i \in I\}$ be a family of E -valued random variables on $(\Omega, \mathbb{P})$, indexed by some countable set I , and let $\{a_n \mid n \in \mathbb{N}\}$ be an I -indexed family of real numbers. Prove that

$$\mathbb{E} \left[\sum_{i \in I} a_i X_i \right] = \sum_{i \in I} a_i \mathbb{E}[X_i]$$

◁

Section 11.E

Chapter 11 exercises

True–False questions

In Questions 11.E.1 to 11.E.5, determine (with proof) whether the statement is true or false.

11.E.1. In a probability space $(\Omega, \mathbb{P})$, we have $\mathbb{P}(A \mid \Omega) = \mathbb{P}(A)$ for all events A .

11.E.2. The function $\mathbb{P} : \mathcal{P}(\mathbb{N}) \rightarrow [0, 1]$ defined by $\mathbb{P}(A) = \sum_{n \in A} 2^{-n}$ is a probability measure on $\mathbb{N}$.

11.E.3. The function $\mathbb{P} : \mathcal{P}(\mathbb{Z}) \rightarrow [0, 1]$ defined by $\mathbb{P}(A) = \sum_{n \in A} 2^{-n}$ is a probability measure on $\mathbb{Z}$.

11.E.4. The function $\mathbb{P} : \mathcal{P}(\mathbb{N}) \rightarrow [0, 1]$, defined by letting $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(A) = 2^{-\min(A)}$ for all $A \neq \emptyset$, is a probability measure on $\mathbb{N}$.

11.E.5. A random variable can have infinite expectation.

Always–Sometimes–Never questions

In Questions 11.E.6 to 11.E.10, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

11.E.6. Let $(\Omega, \mathbb{P})$ be a probability space. The only event whose probability is zero is the empty event $\emptyset \subseteq \Omega$.

11.E.7. Let $(\Omega, \mathbb{P})$ be a probability space and let A be an event. Then $\mathbb{P}(A)^2 - \mathbb{P}(A) = 0$.

11.E.8. Let $(\Omega, \mathbb{P})$ be a probability space and let A and B be events with $\mathbb{P}(B) > 0$. Then $\mathbb{P}(A \mid B) \geq \mathbb{P}(A)$.

11.E.9. Let $(\Omega, \mathbb{P})$ be a probability space and let A and B be mutually exclusive events with $\mathbb{P}(B) > 0$. Then $\mathbb{P}(A \mid B) = 0$.

11.E.10. Let $(\Omega, \mathbb{P})$ be a probability space and let A , B and C be events with $\mathbb{P}(B) > 0$ and $\mathbb{P}(C) > 0$. Then $\mathbb{P}(A \mid B)\mathbb{P}(B \mid C) = \mathbb{P}(A \mid C)$.

Chapter 12

Additional topics

Section 12.1

Orders and lattices

Exercise 12.1.1

Let X be a set. The poset $(\mathcal{P}(X), \subseteq)$ has a least element and a greatest element; find both. ◁

Exercise 12.1.2

Prove that the least element of $\mathbb{N}$ with respect to divisibility is 1, and the greatest element is 0. ◁

Exercise 12.1.3

Let X be a set, and let $U, V \in \mathcal{P}(X)$. Prove that the $\subseteq$ -supremum of $\{U, V\}$ is $U \cup V$, and the $\subseteq$ -infimum of $\{U, V\}$ is $U \cap V$. ◁

Exercise 12.1.4

Let $a, b \in \mathbb{N}$. Show that $\gcd(a, b)$ is an infimum of $\{a, b\}$ and that $\text{lcm}(a, b)$ is a supremum of $\{a, b\}$ with respect to divisibility. ◁

Exercise 12.1.5 (Properties of suprema and infima)

Let $(X, \preceq)$ be a lattice. Prove that, for all $x, y \in X$, we have:

- (a) (**Idempotence**) $x \wedge x = x$ and $x \vee x = x$;
- (b) (**Commutativity**) $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$;
- (c) (**Absorption**) $x \vee (x \wedge y) = x$ and $x \wedge (x \vee y) = x$.

Exercise 12.1.6

Prove that $(\mathbb{N}, |)$ is a distributive lattice. ◁

Exercise 12.1.7

Let $(X, \preceq)$ be a distributive lattice with a greatest element and a least element, and let $x \in X$. Prove that, if a complement for x exists, then it is unique; that is, prove that if $y, z \in X$ are complements for x , then $y = z$. ◁

 **Exercise 12.1.8**
Prove part (b) of ??.



Section 12.2

Inductively defined sets

 Exercise 12.2.1

Fix an alphabet Σ . Following ??, define rules that describe the set Σ^* of words over Σ . How would your rules need to be changed if the empty word ε were not allowed? ◁

 Exercise 12.2.2

Let $\Sigma = \{a, b, c, d\}$. Draw a diagram to represent how the word $cbadbd \in \Sigma^*$ can be obtained by applying the rules you defined in [Exercise 12.2.1](#). ◁

 Exercise 12.2.3

Let Σ be an alphabet. Prove that Σ^* is inductively defined by the rules $(\mid \varepsilon)$ and $\sigma_a = (w \mid wa)$ for $a \in \Sigma$. ◁

 Exercise 12.2.4

Prove that $\mathbb{N}$ is inductively defined by the rules $(\mid 0)$, $(\mid 1)$ and $(n \mid n + 2)$. ◁

 Exercise 12.2.5

Let R be a set of rules for an inductive definition and let A be the set generated by R . Prove that if R has no nullary rules, then A is empty. ◁

 Exercise 12.2.6

Let R be a set of rules for an inductive definition, and let A be the set generated by R . Prove that if R is countable, then A is countable. ◁

 Exercise 12.2.7

Let R be a set of rules. Prove that the set A generated by R is inductively defined; for each rule $\sigma = (\mathbf{x} \mid \sigma(\mathbf{x}))$, the constructor $f_\sigma : A^{\text{ar}(\sigma)} \rightarrow A$ is defined by

$$f_\sigma(\mathbf{a}) = \sigma(\mathbf{a})$$

where $\sigma(\mathbf{a})$ denotes the result of substituting the specific elements of A listed in $\mathbf{a}$ for the respective variables listed in $\mathbf{x}$. [For example, if $\sigma = (x, y \mid [x \odot y])$ and $\mathbf{a} = (2, 5)$, then $\sigma(2, 5) = [2 \odot 5]$.] ◁

 Exercise 12.2.8

Let Σ be an alphabet and let Σ^* be the inductively defined set of words over Σ . Prove that for all $w \in \Sigma$, the rank $\text{rk}(w)$ is equal to the length of W . [We regard the empty string ε to have length 0, despite the fact that the placeholder character ε is used to denote it.] ◁

 Exercise 12.2.9

Let P be a set of propositional variables. Prove by structural induction that the rank of a propositional formula $\varphi \in \mathcal{L}(P)$ is equal to the number of logical operators in φ . ◁

 Exercise 12.2.10

Prove that the set $\mathbb{Z}^+$ of all positive integers is a quotient-inductive set given by the rules $(\mid 1)$ and $(x \mid x \cdot p)$ for each positive prime $p \in \mathbb{Z}$. Describe the corresponding inductively defined set A and the quotient map $q : A \rightarrow \mathbb{Z}^+$ explicitly. ◁

Chapter 12 exercises

Appendices

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Appendix A

Proof-writing

Section A.1

Elements of proof-writing

Exercise A.1.1

Consider the following proof that the product of two odd integers is odd.

$$a, b \in \mathbb{Z} \text{ both odd} \Rightarrow a = 2k + 1, b = 2\ell + 1, k, \ell \in \mathbb{Z}$$

$$ab = (2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell + 1 = 2(2k\ell + k + \ell) + 1$$

$$2k\ell + k + \ell \in \mathbb{Z} \therefore ab \text{ odd}$$

Rewrite the proof so that it is easier to read it as prose. Read it aloud, and transcribe what you read. 

Exercise A.1.2

Consider the following notation-heavy proof that $X \cap (Y \cup Z) \subseteq (X \cap Y) \cup (X \cap Z)$ for all sets X , Y and Z .

Let X , Y and Z be sets and let $a \in X \cap (Y \cup Z)$. Then

$$a \in X \wedge a \in Y \cup Z \Rightarrow a \in X \wedge (a \in Y \vee a \in Z)$$

- **Case 1:** $a \in Y \Rightarrow (a \in X \wedge a \in Y) \Rightarrow a \in X \cap Y \Rightarrow a \in (X \cap Y) \cup (X \cap Z)$.
 - **Case 2:** $a \in Z \Rightarrow (a \in X \wedge a \in Z) \Rightarrow a \in X \cap Z \Rightarrow a \in (X \cap Y) \cup (X \cap Z)$.
- $\therefore \forall a, a \in X \cap (Y \cup Z) \Rightarrow a \in (X \cap Y) \cup (X \cap Z)$
 $\therefore X \cap (Y \cup Z) \subseteq (X \cap Y) \cup (X \cap Z)$.

Read the proof aloud and transcribe what you said. Then rewrite the proof with a more appropriate balance of notation and text. 

Section A.2

Vocabulary for proofs **Exercise A.2.1**

In the following proof that every multiple of four is even, identify all instances where an assumption is stated or cited in order to justify a step in the proof.

| Let $n \in \mathbb{Z}$ and suppose that n is divisible by 4. Then by definition of divisibility we have
| $n = 4k$ for some $k \in \mathbb{Z}$. But then n is even, since $n = 4k = 2(2k)$ and $2k \in \mathbb{Z}$.

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Appendix B

Mathematical miscellany

Section B.1

Set theoretic foundations

 Exercise B.1.1

Use the axiom of foundation to prove that there is no sequence of sets $X_0, X_1, X_2, \dots$ such that $X_{n+1} \in X_n$ for all $n \in \mathbb{N}$. ◁

 Exercise B.1.2

Prove that the axioms of infinity and separation imply the existence of an empty set. ◁

Section B.2

Constructions of the number sets

 Exercise B.2.1

Write out the elements of $3_{\mathbb{N}}$ ($= \emptyset^{+++}$) and of $4_{\mathbb{N}}$. ◁

 Exercise B.2.2

Recall the definition of von Neumann natural numbers from ?? . Prove that $|n_{\mathbb{N}}| = n$ for all $n \in \mathbb{N}$. ◁

 Exercise B.2.3 (Zermelo natural numbers)

Set $0_{\mathbb{Z}} = \emptyset$, and for all objects x define $s_{\mathbb{Z}}(x) = \{x\}$. Let

$$\mathbb{N}_{\mathbb{Z}} = \{0_{\mathbb{Z}}, s_{\mathbb{Z}}(0_{\mathbb{Z}}), s_{\mathbb{Z}}(s_{\mathbb{Z}}(0_{\mathbb{Z}})), s_{\mathbb{Z}}(s_{\mathbb{Z}}(s_{\mathbb{Z}}(0_{\mathbb{Z}}))), \dots\}$$

Prove that $(\mathbb{N}_{\mathbb{Z}}, 0_{\mathbb{Z}}, s_{\mathbb{Z}})$ is a notion of natural numbers; this construction of the natural numbers is called the **Zermelo construction**. ◁

 Exercise B.2.4 (Natural numbers with an arbitrary base)

Let $b \in \mathbb{Z}$, and define $\mathbb{Z}_{\geq b} = \{n \in \mathbb{Z} \mid n \geq b\}$; for example, $\mathbb{Z}_{\geq 1} = \{1, 2, 3, 4, 5, \dots\}$. Let $s_b : \mathbb{Z}_{\geq b} \rightarrow \mathbb{Z}_{\geq b}$ be the function defined by $s_b(n) = n + 1$ for all $n \in \mathbb{Z}_{\geq b}$.

Prove that $(\mathbb{Z}_{\geq b}, b, s_b)$ is a notion of natural numbers, and find an explicit description of the unique bijection $f : \mathbb{N}_{\mathbb{N}} \rightarrow \mathbb{Z}$ described in the statement of ?? . ◁

 Exercise B.2.5

Prove that the recursive definition of $\leq$ is equivalent to the definition of ' $m \leq n$ ' as ' $\exists k \in \mathbb{N}, m + k = n$ ', provided we use the convention that $\mathbb{N} = \{0, 1, 2, 3, \dots\}$. ◁

 Exercise B.2.6

Prove that every element of $\mathbb{Z}$, as defined in ?? , is equal to exactly one of the following: $[(0, 0)]_{\sim}$, or $[(0, n)]_{\sim}$ for some $n > 0$, or $[(n, 0)]_{\sim}$ for some $n > 0$. ◁

 Exercise B.2.7

Prove that the function $i : \mathbb{Z} \rightarrow \mathbb{Q}$ defined by $i(a) = \frac{a}{1}$ is an injection, and for all $a, b \in \mathbb{Z}$ we have $i(a + b) = i(a) + i(b)$, $i(ab) = i(a)i(b)$, $i(-a) = -i(a)$ and $i(a - b) = i(a) - i(b)$. ◁

 Exercise B.2.8

Let $n > 0$ be composite. Prove that $\mathbb{Z}/n\mathbb{Z}$ is not a field, where zero, unit, addition and multiplication are defined as in ?? . ◁

 Exercise B.2.9

Let $(X, 0, 1, +, \cdot)$ be a field. Prove that $-(-x) = x$ for all $x \in X$, and that $(x^{-1})^{-1} = x$ for all nonzero $x \in X$. ◁

 Exercise B.2.10

Let $(X, 0, 1, +, \cdot)$ be a field. Prove that $(-1) \cdot x = -x$ for all $x \in X$, and that $(-x)^{-1} = -(x^{-1})$ for all nonzero $x \in X$. ◁

 Exercise B.2.11

Let $(X, 0, 1, +, \cdot)$ be a field. Prove that if X is finite, then there is no relation $\leq$ on X such that $(X, 0, 1, +, \cdot, \leq)$ is an ordered field. ◁

Section B.3

Limits of functions **Exercise B.3.1**

Let $a, b \in \mathbb{R}$ with $a < b$. Prove that $\overline{(a, b)} = \overline{(a, b]} = \overline{[a, b)} = [a, b]$. $\triangleleft$

 Exercise B.3.2

Let $f : D \rightarrow \mathbb{R}$ be a function, let $a \in D$ and let $\ell \in \mathbb{R}$. Fix some sequence (x_n) of elements of D , not eventually constant, such that $(x_n) \rightarrow a$. Prove that if $f(x) \rightarrow \ell$ as $x \rightarrow a$, then the sequence $(f(x_n))$ converges to ℓ . $\triangleleft$

 Exercise B.3.3

Let $f : D \rightarrow \mathbb{R}$ be a function, let $a \in D$, and $\ell_1, \ell_2 \in \mathbb{R}$. Prove that if $f(x) \rightarrow \ell_1$ as $x \rightarrow a$, and $f(x) \rightarrow \ell_2$ as $x \rightarrow a$, then $\ell_1 = \ell_2$. $\triangleleft$

 Exercise B.3.4

Let $f : D \rightarrow \mathbb{R}$ be a function and $\ell_1, \ell_2 \in \mathbb{R}$. Prove that if D is unbounded above, and if $f(x) \rightarrow \ell_1$ as $x \rightarrow \infty$ and $f(x) \rightarrow \ell_2$ as $x \rightarrow \infty$, then $\ell_1 = \ell_2$. Prove the analogous result for limits as $x \rightarrow -\infty$ in the case when D is unbounded below. $\triangleleft$

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